

Deep Learning for VWAP Execution in Crypto Markets: Beyond the Volume Curve

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Abstract

Volume-Weighted Average Price (VWAP) is arguably the most prevalent benchmark for trade execution as it provides an unbiased standard for comparing performance across market participants. However, achieving VWAP is inherently challenging due to its dependence on two dynamic factors—volumes and prices. Traditional approaches typically focus on forecasting the market’s volume curve, an assumption that may hold true under steady conditions but becomes suboptimal in more volatile environments or markets such as cryptocurrency where prediction error margins are higher. In this study, I propose a deep learning framework that directly optimizes the VWAP execution objective by bypassing the intermediate step of volume curve prediction. Leveraging automatic differentiation and custom loss functions, my method calibrates order allocation to minimize VWAP slippage, thereby fully addressing the complexities of the execution problem. My results demonstrate that this direct optimization approach consistently achieves lower VWAP slippage compared to conventional methods, even when utilizing a naive linear model presented in [9]. They validate the observation that strategies optimized for VWAP performance tend to diverge from accurate volume curve predictions and thus underscore the advantage of directly modeling the execution objective. This research contributes a more efficient and robust framework for VWAP execution in volatile markets, illustrating the potential of deep learning in complex financial systems where direct objective optimization is crucial. Although my empirical analysis focuses on cryptocurrency markets, the underlying principles of the framework are readily applicable to other asset classes such as equities.

1 Introduction

In a rapidly evolving financial landscape, Volume-Weighted Average Price (VWAP) strategies have become a cornerstone for executing high-volume trades, as they aim to minimize market impact and reduce execution costs. VWAP is widely regarded for its fair and neutral calibration, making it an industry standard for comparing performance across market participants [1]. With the rise of systematic trading, execution strategies have received increasing attention; more specifically, significant institutional trades are now executed using algorithms [21]. Despite this, academic research has predominantly focused on Implementation Shortfall (IS) [26] strategies, while VWAP, despite its extensive industrial use and reputation as a robust benchmark (particularly for large orders) [22], has not been as thoroughly explored. The objective of this study is to propose an alternative VWAP execution approach that moves beyond methods relying solely on forecasting the market volume curve. Indeed, while early approaches in the literature were able to incorporate the price–volume relationship under simplified modeling assumptions, more recent works tend to focus exclusively on the volume curve. This shift arises because more realistic and higher-performing volume models render the simultaneous representation of the volume–volatility (and hence price–volume) relationship considerably more complex. The purpose of this paper is to introduce a method that easily overcomes this limitation, thereby more accurately capturing the full VWAP decision problem. To achieve this, I leverage deep learning as a generalized function calibrator. This approach allows the direct representation of the execution objective in a single step—bypassing the conventional two-step process of first predicting the volume curve and then allocating orders. Traditional econometric and machine learning tools typically rely on this two-step resolution, which can be suboptimal in the presence of prediction errors. The primary challenge in VWAP execution is to optimally allocate trading volumes over the execution horizon—a problem for which historical data does not offer a direct solution. Existing literature often simplifies this challenge by assuming that matching the market’s volume curve is ideal. However, this assumption overlooks the fact that perfect volume predictions are unattainable, especially in the volatile cryptocurrency market. Recognizing and incorporating these inevitable prediction errors into the allocation strategy is essential for achieving a price closer to the market VWAP. Moreover, while conventional models can be calibrated to forecast the volume curve from historical data, directly calibrating an execution strategy is not straightforward because the optimal allocation is unknown in advance. Deep learning offers a significant advantage here. Unlike frameworks that minimize the error between predicted and observed data points, deep learning enables the implementation and optimization of any loss function—thanks to automatic differentiation. The central idea of this paper is to build a model that ingests market data, produces an allocation curve for the order, and is trained to minimize the absolute or quadratic error between the achieved execution price and the market VWAP. Finally, although some studies propose dynamic approaches that update predictions throughout an order’s lifetime, I opt for a static framework for two reasons. First, I aim to demonstrate that even a relatively simple network—such as the naive linear model I use—can enhance performance when directly

representing the trading strategy. Second, incorporating real-time dynamics introduces additional technological complexities that are beyond the scope of this study. My goal is to propose a simple yet effective method for VWAP execution that fully captures the decision problem and leverages deep learning’s flexibility to directly optimize the trader’s true objective.

2 Related work

One of the earliest works on VWAP strategies is by [19], who proposed an optimal slicing strategy for VWAP execution. In scenarios where volume and volatility are uncorrelated, the optimal execution curve aligns with the relative market volume curve; deviations in the correlated case were also quantified. Building on this, [23] extended Konishi’s model by incorporating a constrained trading rate and an additional drift term—reflecting situations where traders or brokers, armed with sensitive information, may attempt to “beat” the VWAP. Their study also derived stylized facts regarding the expected relative volume (normalized by trade count), observing an S-shaped pattern throughout the trading day for equities, with higher-turnover stocks exhibiting less variability. These patterns echo the well-known U-shaped trading activity in equity markets. Later, [24] advanced this work by developing a continuously dynamic model, demonstrating that an optimal VWAP trading strategy is closely linked to intraday volume estimation. Volumes have also been used as covariates to explain other market variables such as price or volatility [6, 28, 27, 17], typically using lower-frequency data rather than intraday figures. Additionally, studies such as [12] have focused on measuring overall market trading activity through volume metrics.

In 2008, [2] proposed a method for estimating intraday volumes by decomposing them into two components—an approach refined in [3] by separating volumes into a market evolution component and a stock-specific pattern. The dynamic component was modeled using ARMA and SETAR techniques, resulting in significantly improved accuracy over static volume curve approaches. However, while earlier works like [19] and [23] attempted to capture the interrelation between price and volume within relatively simple frameworks, [2] and [3] focused exclusively on the volume component, as incorporating a realistic volume–price relationship renders overall modeling significantly more complex. An alternative execution method is presented in [14], where Dynamic VWAP (DVWAP) is introduced in contrast to the conventional Historical VWAP (HVWAP). It is argued that incoming news during execution can affect volumes in both directions—information that HVWAP ignores. Their DVWAP framework, built upon earlier methodologies (e.g., those of Bialkowski, Darolles, and Le Fol), not only leverages these improvements but further enhances VWAP performance. Other approaches include stochastic methods proposed by [4] and [7]. As noted by Frei and Westray, these models derive an optimal trading rate that depends solely on volume curves, neglecting price dynamics due to the assumption that the price process behaves as an uncorrelated Brownian motion. In [5], the focus shifts partially to the strategic aspects of VWAP execution at high frequencies, addressing the dilemma faced by brokers between using aggressive versus passive orders. Finally, [13] introduced

two novel contributions. First, they integrated both temporary and permanent market impact components—an important consideration for large institutional orders aiming to minimize market impact. Second, they proposed a method for pricing guaranteed VWAP services using a CARA utility function for the broker and indifference pricing. Unlike other approaches that solely strive for benchmark proximity, their framework emphasizes achieving optimal execution while managing risk. A common theme among these approaches is the predominant focus on modeling market volumes, often under the assumption that prices and volumes are independent—a simplification that does not fully capture market reality. Until recently, the latest advancements in machine learning—particularly deep learning—had not been fully leveraged for VWAP execution. Recent studies, however, have begun to explore these novel methods. For instance, [20] proposed the Macro-Meta-Micro Trader (M3T) architecture, which combines deep learning with hierarchical reinforcement learning to capture market patterns and execute orders across multiple temporal scales. Although this approach achieved an average cost saving of 1.16 basis points relative to an optimal baseline, its complexity may hinder practical implementation. Similarly, [18] developed a dual-level reinforcement learning strategy using Proximal Policy Optimization (PPO) to track daily cumulative VWAP. Their method integrates a Transformer model to capture the overall U-shaped volume pattern with an LSTM model for finer order distribution. Despite improved accuracy over previous reinforcement learning models, it remains heavily reliant on volume prediction. In [25], large stock order execution is simulated using LSTMs within the Almgren and Chriss framework. By leveraging cross-sectional data from multiple stocks to capture inter-stock dependencies, their approach consistently outperforms TWAP and VWAP-based strategies on S&P100 data. However, the focus remains on minimizing transaction costs rather than directly optimizing VWAP execution. While recent studies underscore the potential of machine learning techniques for VWAP execution, many of these methods continue to depend on traditional volume curve prediction rather than directly targeting the VWAP execution objective.

3 Defining the VWAP Problem

The *Volume Weighted Average Price (VWAP)* is a trading benchmark that reflects the average price at which a security is traded throughout the day, weighted by volume. It provides insights into both the trend and the value of a security.

- Its basic formulation is:

$$\text{VWAP} = \frac{\text{Total Traded Value}}{\text{Total Traded Volume}}. \quad (1)$$

- More explicitly, VWAP is computed as:

$$\text{VWAP} = \frac{\sum_i (p_i \times v_i)}{\sum_i v_i}, \quad (2)$$

where p_i denotes the price of the i th trade and v_i the corresponding volume.

Defining the Execution Problem

The objective is to execute trades so that the average execution price aligns as closely as possible with the market VWAP over a given period T . The total order volume is denoted by v and the total market volume over the same period by V .

The average execution price is defined as:

$$P = \frac{\sum_{j=1}^n p_j \times v_j}{v}, \quad \text{with } v = \sum_{j=1}^n v_j, \quad (3)$$

assuming that n transactions are executed during period T .

Similarly, the market VWAP is defined as:

$$\text{VWAP} = \frac{\sum_{i=1}^N p_i \times V_i}{V}, \quad \text{with } V = \sum_{i=1}^N V_i, \quad (4)$$

where N represents the total number of market transactions (including the order's own trades).

Addressing Slippage

Absolute slippage is defined as:

$$S = |P - \text{VWAP}|. \quad (5)$$

Minimizing S is challenging because the transaction space is non-standard—trades occur as discrete events at irregular intervals rather than in uniform time or volume increments. To simplify the analysis, the overall period is divided into T smaller intervals. Within each interval, the following definitions apply:

$$\text{VWAP}_T = \frac{\sum_{t=1}^T \text{VWAP}_t \times V_t}{V}, \quad P_T = \frac{\sum_{t=1}^T P_t \times v_t}{v}, \quad (6)$$

where:

- V_t is the market volume in interval t ,
- v_t is the executed volume in interval t ,
- VWAP_t and P_t denote the market VWAP and the average execution price in interval t , respectively.

This segmentation permits a more detailed comparison between P_T and VWAP_T .

Refining the Problem Definition

For further analysis, normalized volume proportions are introduced:

$$Q_t = \frac{V_t}{V} \quad \text{and} \quad q_t = \frac{v_t}{v},$$

so that:

$$\sum_{t=1}^T Q_t = \sum_{t=1}^T q_t = 1.$$

Thus, the execution price and market VWAP can be rewritten as:

$$P_T = \sum_{t=1}^T P_t q_t \quad \text{and} \quad \text{VWAP}_T = \sum_{t=1}^T \text{VWAP}_t Q_t.$$

Accordingly, the slippage becomes:

$$S_T = |P_T - \text{VWAP}_T| = \left| \sum_{t=1}^T (P_t q_t - \text{VWAP}_t Q_t) \right|. \quad (7)$$

A common strategy to decompose the difference is to add and subtract a common term inside the summation. In this case, $\text{VWAP}_t q_t$ is added and subtracted for each interval:

$$P_t q_t - \text{VWAP}_t Q_t = \underbrace{(P_t q_t - \text{VWAP}_t q_t)}_{=(P_t - \text{VWAP}_t)q_t} + \underbrace{(\text{VWAP}_t q_t - \text{VWAP}_t Q_t)}_{=\text{VWAP}_t (q_t - Q_t)}. \quad (8)$$

Substituting this into (7) yields:

$$S_T = \left| \sum_{t=1}^T [(P_t - \text{VWAP}_t)q_t + \text{VWAP}_t (q_t - Q_t)] \right|. \quad (9)$$

Since the absolute value of a sum is generally not equal to the sum of the absolute values, the triangle inequality is invoked:

$$\left| \sum_{t=1}^T a_t \right| \leq \sum_{t=1}^T |a_t|.$$

Thus,

$$S_T \leq \sum_{t=1}^T |(P_t - \text{VWAP}_t)q_t + \text{VWAP}_t (q_t - Q_t)|. \quad (10)$$

Furthermore, applying the triangle inequality to each individual term results in:

$$\begin{aligned} |(P_t - \text{VWAP}_t)q_t + \text{VWAP}_t (q_t - Q_t)| \\ \leq |(P_t - \text{VWAP}_t)q_t| + |\text{VWAP}_t (q_t - Q_t)|. \end{aligned} \quad (11)$$

Summing over t gives:

$$S_T \leq \sum_{t=1}^T |(P_t - \text{VWAP}_t)q_t| + \sum_{t=1}^T |\text{VWAP}_t(q_t - Q_t)|. \quad (12)$$

In this decomposition:

- $|(P_t - \text{VWAP}_t)q_t|$ represents the deviation in price weighted by the executed volume proportion.
- $|\text{VWAP}_t(q_t - Q_t)|$ captures the impact of discrepancies between the executed volume allocation and the market's volume distribution.

Methodology — Delineating the Problems

In the pursuit of minimizing slippage, two distinct challenges are identified.

Problem 1: Price Deviation Minimization

This aspect focuses on minimizing:

$$\sum_{t=1}^T |(P_t - \text{VWAP}_t)q_t|.$$

Key considerations include:

- Execution quality within each interval is paramount; the goal is to achieve prices as close as possible to the market VWAP.
- With finer time intervals, price deviations are expected to be smaller, and deviations across bins may partially offset one another.
- This challenge is primarily related to market microstructure and the quality of executions.

Problem 2: Volume Allocation Optimization

This challenge is characterized by:

$$\sum_{t=1}^T |\text{VWAP}_t(q_t - Q_t)|. \quad (13)$$

Key considerations include:

- Although VWAP_t cannot be controlled, the executed volume allocation q_t can be adjusted.

- Traditional methods focus on accurately predicting Q_t , the market volume profile; however, given the inherent uncertainty in market dynamics, this may not be optimal.
- The strategy is to design a q_t allocation that accounts for the noisy nature of the Q_t process, adapting to market conditions and prediction errors.
- For instance, during periods of low volatility, it may be advisable to execute less than the market proportion Q_t to mitigate the risk of sudden volatility spikes.
- The aim is to develop a robust volume allocation strategy that minimizes slippage even when predictions of Q_t are imperfect.

Choosing the Appropriate Focus

While the primary aim is to minimize absolute slippage, it is also essential to consider strategies that not only meet the benchmark but also potentially outperform it.

Price Difference Focus

- Since the weights q_t and Q_t are non-negative, achieving the most favorable prices in each interval is crucial.
- Opportunities for value addition may arise, particularly through the mitigation of trading costs such as the spread.

Volume Allocation Focus

- Given the challenges in precisely forecasting $VWAP_t$, emphasis is placed on risk reduction through effective volume allocation.
- The strategy aims to minimize the term $|VWAP_t (q_t - Q_t)|$ by aligning the executed volume with the market's volume profile.
- This involves leveraging insights into future market volumes and understanding their interaction with market volatility.
- The focus is on developing the optimal q_t strategy for volume allocation over larger time intervals, thereby reducing the risk associated with VWAP execution.

In summary, by rigorously decomposing the slippage S_T and carefully applying the triangle inequality, the problem is separated into two distinct components: execution quality (price differences) and volume allocation discrepancies. This detailed derivation provides a solid foundation for subsequent analysis and strategy development.

4 A fixed optimal allocation curve approach

To demonstrate the validity of allocation strategies that are not based solely on following the volume curve, an analysis was conducted in which an optimal allocation curve was calibrated using only historical data, without incorporating current market information. In this experiment, the allocation curve is represented by a simple vector of weights that determines the fraction of the total order executed in each time bin. Without any additional market inputs, the expected volume curve is a uniform distribution over the execution period—precisely as conventional wisdom would predict—so any deviation from this flat pattern in the optimal allocation would challenge that notion.

4.1 Methodology

Historical data from Binance perpetual futures contracts for Bitcoin (BTC), Ethereum (ETH), and Cardano (ADA) covering the period from contract inception until July 1, 2024, were employed. The data, recorded at an hourly frequency, were divided into training and testing sets in an 80/20 split, with the temporal order preserved to ensure a realistic evaluation. The VWAP execution problem was formulated as the optimization of a vector of allocation weights, q_t (for $t = 1, \dots, T$ with $T = 8$ bins), by minimizing one of three loss functions. The first is the Quadratic VWAP Loss, defined as

$$L_Q = \mathbb{E} \left[\left(\frac{\text{VWAP}_{\text{achieved}}}{\text{VWAP}_{\text{market}}} - 1 \right)^2 \right], \quad (14)$$

the second is the Absolute VWAP Loss,

$$L_A = \mathbb{E} \left[\left| \frac{\text{VWAP}_{\text{achieved}}}{\text{VWAP}_{\text{market}}} - 1 \right| \right], \quad (15)$$

and the third is the Volume Curve Loss,

$$L_V = \mathbb{E} \left[\sum_{t=1}^T \left(\frac{v_t}{\sum_{s=1}^T v_s} - \frac{V_t}{\sum_{s=1}^T V_s} \right)^2 \right]. \quad (16)$$

The achieved and market VWAP are defined respectively as

$$\text{VWAP}_{\text{achieved}} = \frac{\sum_{t=1}^T v_t p_t}{\sum_{t=1}^T v_t}, \quad (17)$$

$$\text{VWAP}_{\text{market}} = \frac{\sum_{t=1}^T V_t p_t}{\sum_{t=1}^T V_t}. \quad (18)$$

Here, v_t represents the allocated volume in bin t , V_t is the market volume in bin t , and p_t is the VWAP price in bin t . To solve the optimization problem, three different methods were employed. Sequential Least Squares Programming (SLSQP) was used as a local

optimization technique with 1000 restarts from random initial guesses to mitigate the risk of converging to local minima. Basin-Hopping, a global optimization algorithm that combines local searches with random perturbations, was also employed and performed 10 times. In addition, Differential Evolution, an evolutionary algorithm known for its robustness in finding global optima, was executed once. Performance metrics for each asset, loss function, and optimization method are summarized in Table 1, while Figure 1 illustrates the optimal allocation curves derived from these optimizations.

Table 1: VWAP Optimization Results

Asset	Optimization Function	Method	Abs VWAP Loss (1e2)	Quad VWAP Loss (1e4)	Vol Curve Loss (1e2)	R2 Vol Curve	Opt Time (s)
ADA	absolute_vwap_loss	SLSQP	0.161262	0.103157	0.509109	-0.347896	34.626976
		basinhopping	0.161266	0.099737	0.556507	-0.473385	55.531589
		DE	0.159714	0.103534	0.463983	-0.228422	2.072001
	quadratic_vwap_loss	SLSQP	0.161468	0.097612	0.615348	-0.629170	11.889972
		basinhopping	0.187822	0.102028	0.963292	-1.550373	27.556498
		DE	0.164090	0.092467	0.576554	-0.526461	3.021690
	volume_curve_loss	SLSQP	0.182052	0.153098	0.379948	-0.005934	25.458261
		basinhopping	0.184498	0.156269	0.382820	-0.013539	36.394290
		DE	0.182067	0.152669	0.377981	-0.000728	1.807036
BTC	absolute_vwap_loss	SLSQP	0.105354	0.037911	0.855709	-0.628073	34.237416
		basinhopping	0.104696	0.037538	0.664384	-0.264057	48.594292
		DE	0.105952	0.040566	0.636615	-0.211224	2.709536
	quadratic_vwap_loss	SLSQP	0.105112	0.037638	0.728556	-0.386152	12.000814
		basinhopping	0.108195	0.037482	1.308381	-1.489326	26.424249
		DE	0.103572	0.035020	0.780900	-0.485740	4.301013
	volume_curve_loss	SLSQP	0.126713	0.061470	0.528446	-0.005422	21.426937
		basinhopping	0.122180	0.057882	0.535264	-0.018393	34.317264
		DE	0.124038	0.058830	0.525396	0.000381	1.163200
ETH	absolute_vwap_loss	SLSQP	0.121005	0.053397	0.696746	-0.418203	36.342283
		basinhopping	0.125103	0.059826	0.680083	-0.384285	52.675163
		DE	0.120213	0.053212	0.622947	-0.267986	2.533839
	quadratic_vwap_loss	SLSQP	0.120978	0.052524	0.709165	-0.443481	11.941791
		basinhopping	0.128118	0.052153	0.881157	-0.793564	27.980689
		DE	0.119349	0.049538	0.678544	-0.381152	3.884477
	volume_curve_loss	SLSQP	0.147338	0.084255	0.493581	-0.004668	25.072747
		basinhopping	0.143035	0.080341	0.491345	-0.000117	33.726844
		DE	0.144048	0.081395	0.491395	-0.000218	2.017230

Note: Abs VWAP Loss and Vol Curve Loss are presented in 10^2 scale, Quad VWAP Loss is presented in 10^4 scale as per the original data.

The analysis reveals several important findings. Allocation curves obtained by minimizing the absolute and quadratic VWAP losses clearly deviate from a uniform allocation, while the optimization based on the volume curve loss yields a nearly flat allocation. This flat allocation is expected since, in the absence of market information, the average volume in each time bin is equal. However, although the flat allocation perfectly tracks the average volume curve (as evidenced by an R^2 near zero), it performs worse in terms of VWAP execution compared to the non-uniform allocations derived from the VWAP-based losses. Furthermore, among the assets studied, Bitcoin consistently exhibits the lowest VWAP losses, reflecting its relative stability, whereas Cardano—being more volatile—shows higher losses. In terms of optimization methods, Differential Evolution consistently produced competitive results across assets and loss functions, while SLSQP, despite numerous restarts, often converged to suboptimal solutions due to the complexity of the optimization landscape. Basin-Hopping displayed variable performance that was

generally less consistent.

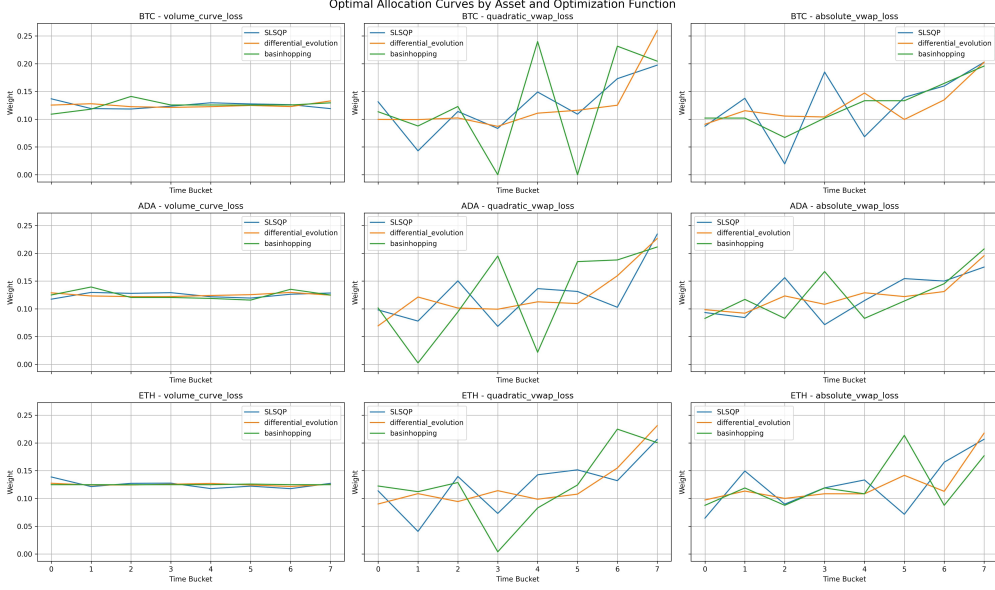


Figure 1: Optimal Allocation based on optimization methods and asset

4.2 Discussion

This experiment is particularly revealing because it involves optimizing a vector of allocation weights using only historical data without current market information. Consequently, an optimization based solely on the volume curve loss produces a flat, uniform allocation, which is expected when no additional information beyond the historical average is available. In contrast, the allocation curves optimized using the absolute and quadratic VWAP losses diverge significantly from uniformity, exhibiting a pronounced end-loading behavior in some cases. This divergence indicates that even in the absence of direct market signals, an allocation strategy that minimizes VWAP slippage is inherently non-uniform. This finding challenges the traditional approach of following the expected volume curve and strongly supports the use of direct VWAP loss optimization in trading execution strategies. Although the empirically derived allocation curves deviate markedly from the conventional volume curve, this outcome is not entirely surprising when considering simplified frameworks that account for price–volume interactions. For instance, Konishi [19] shows that the optimal fraction $x^*(t)$ allocated to bin t can be approximated by

$$x^*(t) \approx \frac{\mathbb{E}[\sigma(t, V)^2 X(t)]}{\mathbb{E}[\sigma(t, V)^2]}, \quad (19)$$

or equivalently, as rewritten by McCulloch and Kazakov [23]:

$$x_t = \frac{\mathbb{E}[X_t \sigma_t^2]}{\mathbb{E}[\sigma_t^2]} = \mathbb{E}[X_t] + \frac{\text{Cov}[X_t, \sigma_t^2]}{\mathbb{E}[\sigma_t^2]}. \quad (20)$$

Thus, a sufficiently large positive covariance term can shift the solution away from the market’s average volume curve, underscoring the role of volatility in shaping optimal allocations. These results highlight the importance of the price–volume relationship and expose the limitations of approaches based solely on replicating the historical volume curve. Although more complex models can be developed to jointly account for both price and volume dynamics, the present analysis demonstrates that even a relatively simple (and naive) linear model can yield meaningful improvements in VWAP execution by directly optimizing the VWAP objective.

5 A Deep-Learning Model for VWAP Allocation

5.1 Framework

Having demonstrated that the traditional volume curve approach may not be optimal for allocating volumes over time, deep learning is employed as a powerful means to represent and optimize the VWAP allocation problem. Deep learning affords the flexibility to directly optimize the specific objective—minimizing the deviation between the achieved execution price and the market VWAP—by leveraging automatic differentiation. In this framework, the model accepts sequential inputs comprising multiple features over a designated historical lookback period and produces an allocation curve for future time intervals. The key innovation is the use of a custom loss function (either absolute or quadratic VWAP loss) that captures the true execution error, thereby enabling the optimization of the strategy in a single step rather than relying on intermediate proxies such as the volume curve. A critical requirement is that the model’s output constitutes a valid allocation strategy; that is, the allocation weights must be non-negative and sum to one. Although several approaches (e.g., clipping negative values or taking absolute values before normalization) were considered, these methods risk introducing uneven optimization. Instead, the softmax function is adopted, which naturally transforms any real-valued vector into a smooth probability distribution over the execution horizon. This ensures that even if the Temporal Linear Network (TLN) produces negative values, the final allocation vector remains both non-negative and normalized, thereby satisfying the constraints without sacrificing optimization smoothness.

5.2 Internal Model

In selecting an internal model for the VWAP allocation framework, various architectures traditionally employed for sequential prediction tasks were considered. Recent studies in volume prediction for cryptocurrency markets have explored models such as Temporal Kolmogorov-Arnold Networks (TKAN) [10], Signature-Weighted Kolmogorov-Arnold Networks (SigKAN) [16], Temporal Kolmogorov-Arnold Transformers (TKAT) [8], Kolmogorov-Arnold Mixture of Experts (KAMoE) [15] and Recurrent Neural Networks with Signature-Based Gating Mechanisms (SigGate) [11]. Although these approaches offer innovative solutions for capturing long-term dependencies and complex interactions, they also introduce additional complexity that is not necessary when the primary goal

is to demonstrate the importance of optimizing the correct objective. Accordingly, the Temporal Linear Network (TLN) [9] was chosen as the default internal model. The TLN is conceptually simple and highly interpretable—essentially a linear model with structured transformations—yet it is integrated within a deep learning framework that enables the minimization of arbitrary loss functions via automatic differentiation. Importantly, even a naive linear model performs effectively within the proposed approach, emphasizing that improvements arise from directly optimizing the VWAP execution objective rather than from architectural complexity. Let the input tensor be

$$X \in \mathbb{R}^{B \times S \times F},$$

where B is the batch size, S is the length of the input sequence, and F is the number of input features. The TLN consists of L layers, each implementing a cascade of linear operations. Each operation within a layer is a linear mapping with learnable weights, and the layers are designed to progressively adjust both the sequence length and feature dimension. This structured design leverages parameter sharing and constraints, resulting in a model that is both computationally efficient and stable.

Each layer ℓ (with $1 \leq \ell \leq L$) transforms its input $X^{(\ell-1)}$ into an output $X^{(\ell)}$ through two primary stages: a **structured linear transformation** and a **convolutional filtering** along the time dimension.

1. Structured Linear Transformation

This stage is composed of three consecutive linear operations.

Temporal Scaling: Each time index is scaled by a learnable vector $K^{(\ell)} \in \mathbb{R}^{S_\ell}$, where S_ℓ denotes the current sequence length at layer ℓ . For each batch index b , time index s , and feature index f , the computation is as follows:

$$\tilde{X}_{b,s,f}^{(\ell)} = K_s^{(\ell)} X_{b,s,f}^{(\ell-1)}. \quad (21)$$

Feature Transformation: At each time step, the features are mapped from $\mathbb{R}^{F_\ell}$ to an intermediate space $\mathbb{R}^{F'_\ell}$ using a learnable weight matrix $W^{(\ell)} \in \mathbb{R}^{F_\ell \times F'_\ell}$ and bias $b^{(\ell)} \in \mathbb{R}^{F'_\ell}$:

$$Y_{b,s,\cdot}^{(\ell)} = \tilde{X}_{b,s,\cdot}^{(\ell)} W^{(\ell)} + b^{(\ell)}. \quad (22)$$

An additional element-wise scaling is then applied using a learnable vector $F^{(\ell)} \in \mathbb{R}^{F'_\ell}$:

$$Z_{b,s,f'}^{(\ell)} = F_{f'}^{(\ell)} \cdot Y_{b,s,f'}^{(\ell)}. \quad (23)$$

Temporal Transformation: Finally, the sequence is remapped to a new length S'_ℓ by applying a linear transformation with learnable parameters $T^{(\ell)} \in \mathbb{R}^{S_\ell \times S'_\ell}$ and bias $c^{(\ell)} \in \mathbb{R}^{S'_\ell}$:

$$\hat{Z}_{b,s'}^{(\ell)} = \sum_{s=1}^{S_\ell} T_{s,s'}^{(\ell)} Z_{b,s,f'}^{(\ell)} + c_{s'}^{(\ell)}. \quad (24)$$

In summary, the structured transformation is represented as

$$\Phi^{(\ell)}(X^{(\ell-1)}) = \mathcal{T}^{(\ell)}\left(\left(X^{(\ell-1)} \odot K^{(\ell)}\right) W^{(\ell)} + b^{(\ell)} \odot F^{(\ell)}\right), \quad (25)$$

where $\mathcal{T}^{(\ell)}(\cdot)$ denotes the temporal mapping and $\odot$ represents element-wise multiplication.

2. Convolutional Filtering

After the structured transformation, each feature channel is convolved along the time dimension. Let $C^{(\ell)} \in \mathbb{R}^{K_{\text{conv}}^{(\ell)} \times F_{\ell}'}$ denote the convolution kernel (with kernel size $K_{\text{conv}}^{(\ell)}$) and $d^{(\ell)} \in \mathbb{R}^{F_{\ell}'}$ the corresponding bias. The convolution is computed as

$$X_{b,t,f'}^{(\ell)} = \sum_{k=0}^{K_{\text{conv}}^{(\ell)}-1} C_{k,f'}^{(\ell)} \Phi^{(\ell)}(X^{(\ell-1)})_{b,t+k,f'} + d_{f'}^{(\ell)}. \quad (26)$$

Thus, the overall operation in layer ℓ is given by

$$X^{(\ell)} = \text{Conv}\left(\Phi^{(\ell)}(X^{(\ell-1)})\right). \quad (27)$$

5.3 The Role of Softmax in Enforcing Allocation Constraints

A key innovation in the framework is the use of the softmax function to enforce allocation constraints. In this context, the final output of the internal model must be a valid allocation vector—that is, all elements must be non-negative and the vector must sum to one. While straightforward normalization (dividing by the sum of the outputs) or clipping negative values might appear sufficient, these approaches can lead to uneven or suboptimal optimization. In contrast, the softmax function

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}, \quad (28)$$

naturally transforms any real-valued input vector z into a smooth probability distribution. This not only satisfies the constraints but also ensures a differentiable transformation that facilitates gradient-based optimization. Even if the TLN produces negative values, the exponential function within softmax guarantees a smooth and balanced allocation across the execution horizon.

5.4 Comparison with Standard Linear Regression and Rationale

A conventional linear regression would flatten the input tensor $X \in \mathbb{R}^{B \times S \times F}$ into $X_{\text{flat}} \in \mathbb{R}^{B \times (S \cdot F)}$ and compute

$$Y = X_{\text{flat}} W_{\text{reg}} + b_{\text{reg}}, \quad (29)$$

resulting in a very large number of parameters. In contrast, the TLN architecture leverages structured, layer-by-layer transformations to progressively adjust the sequence and feature

dimensions, yielding a much more parsimonious model that is less prone to unstable weight estimates. Furthermore, by embedding the TLN within a deep learning framework, it becomes possible to optimize non-traditional loss functions—such as the VWAP-specific losses—using automatic differentiation, an advantage that standard linear regression does not offer.

5.5 Summary of the Optimization Process

The overall optimization process proceeds in three primary steps. First, the TLN processes the input features over a specified lookback period to generate a raw output vector $v_{[t,t+h]} = \text{TLN}(x_t)$ for future time steps. Next, the softmax function is applied to these raw outputs to convert them into a valid allocation vector:

$$q_t = \frac{e^{v_t}}{\sum_{s=1}^T e^{v_s}}, \quad \text{for } t = 1, 2, \dots, T, \quad (30)$$

ensuring that the allocations are non-negative and sum to one. Finally, the model parameters θ are optimized by minimizing the quadratic difference between the achieved VWAP and the market VWAP, as defined in Section 4 (see Eqs. (14)–(18)). In summary, this framework establishes a fair comparison platform by leveraging deep learning to directly optimize the VWAP execution objective. Rather than increasing model complexity, the approach employs a straightforward, linear model (the TLN) enhanced by the softmax transformation to enforce smooth and constrained allocation, demonstrating that aligning the optimization objective precisely with the trading goal can lead to improved execution performance.

6 Results

In this section, a comprehensive comparison of various VWAP execution strategies is presented, including traditional methods, the proposed deep-learning approach, and dynamic strategies. These methods are evaluated across different cryptocurrencies to assess their performance under varying market conditions.

6.1 Benchmarks and Strategies

Several benchmark strategies are considered. As a baseline, a naive flat allocation strategy that uniformly distributes the order volume across all time intervals is included; this simple method serves as a reference for assessing more sophisticated approaches. In addition, fixed optimal allocation curves are evaluated—static allocations optimized using three different loss functions (absolute VWAP loss, quadratic VWAP loss, and volume curve loss) as described in 4. For this optimization, the differential evolution algorithm is employed due to its demonstrated superior and more stable performance. The proposed deep-learning approach, referred to as the *StaticVWAP Model*, employs a neural network to generate allocation curves based on historical market data. For a fair comparison, all

models receive the same input features, including volumes, the hour of the day, the day of the week, and returns computed on the VWAP price of each bin. In contrast, the *Static Linear Regression* model serves as a traditional benchmark that is calibrated to predict the volume. Since linear regression does not inherently produce outputs that satisfy the allocation constraints (i.e., non-negativity and summing to one), its predictions are post-processed by first clipping negative values to zero and then normalizing the resulting vector. In cases where all values are zero (or become zero after clipping), an equiponderated allocation is returned. Dynamic VWAP strategies based on the framework introduced by [3] are also implemented. In these dynamic strategies, the volume executed at each time step is determined by the current market volume prediction and the remaining order size, following the equation

$$v_t = \frac{\hat{V}_t}{\sum_{i=t}^T \hat{V}_i} \cdot \left(1 - \sum_{i=0}^{t-1} v_i\right), \quad (31)$$

where v_t denotes the volume executed at time t , $\hat{V}_t$ is the predicted market volume at time t , T is the end of the execution horizon, and $\sum_{i=0}^{t-1} v_i$ represents the cumulative executed volume up to time $t - 1$. Both a dynamic linear regression approach and a dynamic version of the StaticVWAP model, which updates predictions at each time step, are considered.

6.2 Evaluation Metrics

Three key metrics are used to assess performance. The *Absolute VWAP Loss* measures the absolute difference between the achieved VWAP and the market VWAP, while the *Quadratic VWAP Loss* measures the squared difference, thereby penalizing larger deviations more heavily. In addition, the *R² Score for the Volume Curve* is computed, indicating how accurately the strategy predicts the actual market volume profile.

6.3 Experimental Setup

The strategies are evaluated using hourly data from five major cryptocurrencies (BTC, ETH, BNB, ADA, and XRP) traded on Binance perpetual contracts covering the period from January 1, 2020, to July 1, 2024. This dataset provides sufficient historical data across varying market conditions to evaluate the models' performance in both high and low volatility regimes. The data is partitioned chronologically, with the last 20% reserved as the test set to ensure a true out-of-sample evaluation. A validation set, comprising 20% of the remaining data, is randomly selected from the first 80% of the period to guide model selection and prevent overfitting to any specific market regime.

For input features, several categories are incorporated: (1) Volume data, which is transformed to address non-stationarity by dividing each volume value by a trailing two-week moving average, calculated with appropriate shifting to avoid look-ahead bias; (2) Temporal indicators including hour of day and day of week as categorical features to

capture seasonality patterns; and (3) Price information in the form of returns calculated on the VWAP price of each bin, with a value of 0 assigned when no volume is traded.

The preparation of target variables differs based on the model type. For the neural network approaches, the normalization to volume curve is implemented directly within the loss function, with volumes divided by the sum over the lookback period for numerical stability. For traditional linear regression models, the volume curve is explicitly normalized by dividing each volume by the sum of volumes in the lookahead period, as these models cannot incorporate this normalization within their loss functions.

In these experiments, a lookback period of 120 time steps is used to predict allocations for 12 periods ahead. Additional experiments with modified parameters—one with a 6-step prediction horizon and a shorter lookback period, and another with a 48-step prediction horizon—are provided in the Appendix. The results from these experiments confirm that the framework is effective across different timeframes and that the underlying concepts generalize well.

6.4 Training Details

Standard training settings were adopted. Specifically, the Adam optimizer was employed, and the maximum number of epochs was set to 1000. To mitigate overfitting and ensure stable convergence, two key callbacks were incorporated: an early stopping mechanism, which halts training if no improvement in validation loss is observed for 10 consecutive epochs, and a learning rate reducer, which divides the current learning rate by 4 after 5 epochs without improvement (starting from an initial rate of 0.001). The training set was shuffled, and a 20% validation split was used to monitor these metrics during training.

An additional critical aspect was the mitigation of sensitivity to weight initialization. Each training run was repeated 30 times with different random initializations to compute both the average performance and its standard deviation, thereby providing a robust and statistically sound evaluation of model effectiveness.

6.5 Empirical Results

Table 2 summarizes the performance of various VWAP execution strategies for a 12-step-ahead allocation with a 120-step lookback window. The table reports the mean and standard deviation of the Absolute VWAP Loss (scaled by 10^{-2}), Quadratic VWAP Loss (scaled by 10^{-4}), the R^2 score for the volume curve, and the training time (in seconds) for each strategy, across five assets (BTC, ETH, BNB, ADA, and XRP).

Several key patterns emerge from these results. First, all proposed methods consistently outperform the naive flat allocation approach across all assets, confirming that more sophisticated allocation strategies yield tangible benefits for VWAP execution.

Table 2: VWAP Optimization Results for 12 steps ahead and 120 lookback window

Model Type	Asset	Optimization Function	Abs. VWAP Loss (10^{-2})		Quad. VWAP Loss (10^{-4})		R^2 Vol. Curve		Training Time (s)	
			Mean	Std	Mean	Std	Mean	Std	Mean	Std
Naive	BTC	N/A	0.158743	0.000000	0.087808	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	BTC	Absolute	0.119742	0.000835	0.050175	0.001045	-0.133930	0.038680	17.066662	1.452658
StaticVWAP	BTC	Quadratic	0.120810	0.000759	0.047403	0.001055	-0.361577	0.082153	16.568835	0.591146
StaticVWAP	BTC	Volume	0.149938	0.001896	0.084989	0.001637	0.134904	0.011795	24.140719	3.765817
Dynamic Linear	BTC	N/A	0.142535	0.000000	0.085731	0.000000	0.193757	0.000000	0.241681	0.000000
Static Linear	BTC	N/A	0.146979	0.000000	0.084170	0.000000	0.162770	0.000000	0.244588	0.000000
Dynamic StaticVWAP	BTC	Absolute	0.142334	0.008724	0.076042	0.010107	0.021618	0.038493	15.005514	0.799850
Dynamic StaticVWAP	BTC	Quadratic	0.147608	0.018183	0.074102	0.016007	-0.495363	0.815728	14.517617	0.582770
Dynamic StaticVWAP	BTC	Volume	0.152135	0.004588	0.091675	0.003997	0.148551	0.018475	20.605638	2.945344
Fixed Volume Curve	BTC	Absolute	0.129670	0.000000	0.054213	0.000000	-0.273680	0.000000	17.263421	0.000000
Fixed Volume Curve	BTC	Quadratic	0.127530	0.000000	0.049655	0.000000	-0.467000	0.000000	30.385431	0.000000
Fixed Volume Curve	BTC	Volume	0.160575	0.000000	0.089770	0.000000	-0.003349	0.000000	12.861315	0.000000
Naive	ETH	N/A	0.177758	0.000000	0.116196	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	ETH	Absolute	0.138627	0.000675	0.076506	0.001128	-0.146691	0.026960	16.977891	1.268639
StaticVWAP	ETH	Quadratic	0.139999	0.000451	0.073385	0.001008	-0.297154	0.051581	16.011362	0.614544
StaticVWAP	ETH	Volume	0.170102	0.001785	0.122055	0.001819	0.109135	0.007899	19.888228	3.165192
Dynamic Linear	ETH	N/A	0.158829	0.000000	0.121279	0.000000	0.169994	0.000000	0.249899	0.000000
Static Linear	ETH	N/A	0.166326	0.000000	0.122063	0.000000	0.135824	0.000000	0.278128	0.000000
Dynamic StaticVWAP	ETH	Absolute	0.152641	0.003763	0.093910	0.005712	-0.119677	0.182520	15.180526	1.168761
Dynamic StaticVWAP	ETH	Quadratic	0.162467	0.017372	0.094459	0.013988	-0.633158	0.571764	14.129841	0.686132
Dynamic StaticVWAP	ETH	Volume	0.171854	0.004362	0.127756	0.004184	0.118685	0.018637	17.855957	2.545837
Fixed Volume Curve	ETH	Absolute	0.148407	0.000000	0.076206	0.000000	-0.269217	0.000000	16.648768	0.000000
Fixed Volume Curve	ETH	Quadratic	0.147058	0.000000	0.072852	0.000000	-0.348835	0.000000	23.099389	0.000000
Fixed Volume Curve	ETH	Volume	0.174744	0.000000	0.112633	0.000000	-0.006033	0.000000	9.376649	0.000000
Naive	BNB	N/A	0.174116	0.000000	0.118874	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	BNB	Absolute	0.137741	0.000559	0.076553	0.001233	-0.169736	0.036491	15.882844	0.794736
StaticVWAP	BNB	Quadratic	0.141483	0.000963	0.071090	0.000803	-0.477507	0.071598	15.598018	0.652385
StaticVWAP	BNB	Volume	0.165025	0.001455	0.116136	0.001633	0.086762	0.002815	18.406203	1.898933
Dynamic Linear	BNB	N/A	0.160003	0.000000	0.115190	0.000000	0.127731	0.000000	0.262799	0.000000
Static Linear	BNB	N/A	0.163049	0.000000	0.114673	0.000000	0.101505	0.000000	0.216979	0.000000
Dynamic StaticVWAP	BNB	Absolute	0.146537	0.006959	0.074951	0.002269	-0.434465	0.221577	14.089177	0.626975
Dynamic StaticVWAP	BNB	Quadratic	0.183354	0.040719	0.088591	0.031602	-1.490792	1.373252	13.962810	0.606966
Dynamic StaticVWAP	BNB	Volume	0.165976	0.003899	0.119578	0.003757	0.093674	0.009887	16.622247	1.866124
Fixed Volume Curve	BNB	Absolute	0.146639	0.000000	0.079064	0.000000	-0.283410	0.000000	16.226866	0.000000
Fixed Volume Curve	BNB	Quadratic	0.147920	0.000000	0.074117	0.000000	-0.440821	0.000000	31.472387	0.000000
Fixed Volume Curve	BNB	Volume	0.177473	0.000000	0.122219	0.000000	-0.003080	0.000000	12.545491	0.000000
Naive	ADA	N/A	0.226207	0.000000	0.228809	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	ADA	Absolute	0.187574	0.000640	0.156983	0.004485	-0.201991	0.040232	15.773802	0.727374
StaticVWAP	ADA	Quadratic	0.194378	0.002171	0.139060	0.002399	-0.503548	0.086766	15.721992	0.543023
StaticVWAP	ADA	Volume	0.219882	0.001456	0.244318	0.004173	0.089412	0.004083	18.606664	2.294395
Dynamic Linear	ADA	N/A	0.213691	0.000000	0.258357	0.000000	0.142651	0.000000	0.236307	0.000000
Static Linear	ADA	N/A	0.217345	0.000000	0.253104	0.000000	0.112113	0.000000	0.293910	0.000000
Dynamic StaticVWAP	ADA	Absolute	0.224430	0.018242	0.164912	0.013876	-0.868068	0.389384	14.117266	0.803708
Dynamic StaticVWAP	ADA	Quadratic	0.235337	0.043516	0.168912	0.041709	-1.121577	1.028339	13.869841	0.524214
Dynamic StaticVWAP	ADA	Volume	0.223166	0.005122	0.253597	0.009979	0.094306	0.010872	16.104520	2.195146
Fixed Volume Curve	ADA	Absolute	0.195395	0.000000	0.148712	0.000000	-0.311109	0.000000	15.779793	0.000000
Fixed Volume Curve	ADA	Quadratic	0.199729	0.000000	0.135216	0.000000	-0.573288	0.000000	22.675063	0.000000
Fixed Volume Curve	ADA	Volume	0.231114	0.000000	0.236772	0.000000	-0.005151	0.000000	11.179776	0.000000
Naive	XRP	N/A	0.223855	0.000000	0.275925	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	XRP	Absolute	0.182436	0.000558	0.187474	0.003841	-0.268477	0.054168	16.156652	0.781386
StaticVWAP	XRP	Quadratic	0.188880	0.001564	0.158377	0.003440	-0.800538	0.151472	15.717826	0.651796
StaticVWAP	XRP	Volume	0.218816	0.001646	0.291965	0.003142	0.053552	0.002636	18.000321	2.353914
Dynamic Linear	XRP	N/A	0.213728	0.000000	0.295177	0.000000	0.098195	0.000000	0.290283	0.000000
Static Linear	XRP	N/A	0.216581	0.000000	0.295605	0.000000	0.071538	0.000000	0.248654	0.000000
Dynamic StaticVWAP	XRP	Absolute	0.213859	0.017346	0.169141	0.009132	-1.273806	0.583631	14.198917	0.578075
Dynamic StaticVWAP	XRP	Quadratic	0.249005	0.038793	0.195797	0.028412	-2.392281	1.708965	13.999556	0.594463
Dynamic StaticVWAP	XRP	Volume	0.221568	0.003417	0.299404	0.007929	0.052095	0.005762	15.439939	1.495954
Fixed Volume Curve	XRP	Absolute	0.186341	0.000000	0.178947	0.000000	-0.347910	0.000000	14.544090	0.000000
Fixed Volume Curve	XRP	Quadratic	0.189142	0.000000	0.152440	0.000000	-0.794792	0.000000	23.787354	0.000000
Fixed Volume Curve	XRP	Volume	0.224941	0.000000	0.276570	0.000000	-0.001512	0.000000	12.143459	0.000000

When comparing the methods based on their ability to predict volume curves, we observe that Static Linear Regression is more effective than the StaticVWAP model when the latter is optimized for absolute or quadratic VWAP loss. However, the Dynamic Linear Regression approach consistently achieves the highest R^2 values across all assets, indicating superior volume curve prediction capability. This improved prediction translates into better VWAP execution compared to Static Linear Regression.

Interestingly, despite Dynamic Linear Regression’s advantage in volume prediction, the Fixed Volume Curve approach demonstrates superior VWAP execution performance. This finding underscores a critical insight: accurate volume curve prediction, while beneficial, is not the primary determinant of optimal VWAP execution.

Most significantly, the proposed StaticVWAP deep learning model consistently achieves the best absolute or quadratic VWAP loss values (depending on its optimization target) across all assets. When trained to minimize absolute VWAP loss, it delivers the lowest absolute slippage; similarly, when optimized for quadratic VWAP loss, it produces the lowest quadratic slippage metrics.

Regarding training times, the deep-learning models (StaticVWAP) generally require between 15 and 18 seconds per run, which, although longer than the near-instantaneous training times of linear regression models, represents an acceptable trade-off given the significant performance gains. The dynamic approaches show mixed results when applied to the StaticVWAP model, often degrading performance rather than improving it.

Overall, these results validate the central thesis of this study: directly optimizing the VWAP execution objective—rather than focusing solely on volume curve prediction—is essential for achieving superior execution performance in volatile markets such as cryptocurrencies. The StaticVWAP deep learning framework provides a flexible and effective approach that consistently outperforms conventional methods, even when utilizing a relatively simple model architecture.

6.6 Training Times and Comparison Across Prediction Horizons

Tables 3 and 4 provide additional insight into the practical applicability of each method under different prediction horizons. Several consistent patterns emerge across these timeframes that reinforce the findings from our main experiments.

First, regarding training efficiency, the StaticVWAP models consistently maintain reasonable computational demands across all horizons—typically under 20 seconds—making them practical for real-world applications despite being more complex than linear regression models. Notably, while the fixed volume curve approach converges rapidly for shorter horizons, its training time increases dramatically with longer prediction horizons, eventually requiring several minutes per run for 48-step predictions.

When examining performance across different timeframes, the relative rankings of the models remain remarkably consistent. For both shorter (6-step) and longer (48-step) prediction horizons, the StaticVWAP models optimized directly for VWAP loss consistently outperform both naive approaches and linear regression variants. The fixed volume curve approach remains competitive, particularly when optimized with quadratic loss, but generally does not reach the performance levels of the StaticVWAP model.

Dynamic approaches show consistent patterns across different horizons. Dynamic linear regression demonstrates increasingly strong volume curve prediction capability (higher R^2 values) as the prediction horizon extends, yet this improvement in volume prediction

does not translate to proportional gains in VWAP execution performance. This pattern strongly supports our central thesis that accurate volume prediction, while informative, is not the primary driver of optimal VWAP execution. In fact, the performance gap between methods directly optimized for VWAP execution and those focused on volume prediction tends to persist or even widen with longer horizons.

The stability of these findings across different prediction horizons and assets underscores the robustness of our approach. Whether executing shorter or longer-term VWAP strategies, directly optimizing the execution objective consistently yields superior results compared to approaches that rely primarily on volume curve prediction.

6.7 Limitations and Future Directions

Although the proposed approach yields promising improvements in VWAP execution, several limitations warrant discussion. One key assumption in the current framework is that market impact is negligible. This assumption is common in the VWAP literature, as employing a VWAP benchmark over a long period is believed to mitigate market impact. However, it is recognized that large order sizes may affect execution quality. Studies such as Gueant et al. have shown that both temporary and permanent market impact components can be incorporated into the optimization process, suggesting that future work could integrate such models to more accurately capture real-world trading dynamics. Another limitation is that the present model operates in a static framework without dynamic updating in real time. Although static models have the advantage of simplicity and reduced technological complexity, dynamic adjustments could potentially enhance execution performance in highly volatile environments. Future research may explore hybrid approaches that combine the benefits of static optimization with dynamic market adaptation. Finally, while the analysis in this study focuses on cryptocurrency markets, the underlying principles are readily extendable to other asset classes such as equities. Incorporating more sophisticated market impact models and dynamic updating mechanisms will be valuable in developing a comprehensive framework for VWAP execution across diverse trading environments.

6.8 Understanding the Predicted Curves

An analysis of the predicted allocation curves under different calibration losses is presented to elucidate how the proposed model differentiates itself from competing approaches. Figures 2 and 3 illustrate that the predictions generated using absolute and quadratic VWAP losses tend to follow an average allocation pattern similar to that observed in Section 2, while still adapting to current market conditions. Notably, the model trained with quadratic loss produces a more compact prediction region, suggesting that it sacrifices flexibility in favor of an end-loaded execution pattern. In contrast, the model trained with volume curve loss results in a nearly flat allocation, with distinct seasonal variations clearly visible. These visual differences provide important insight into how the choice of loss function shapes the resulting allocation strategy, ultimately contributing to the performance differences observed in the quantitative analysis.

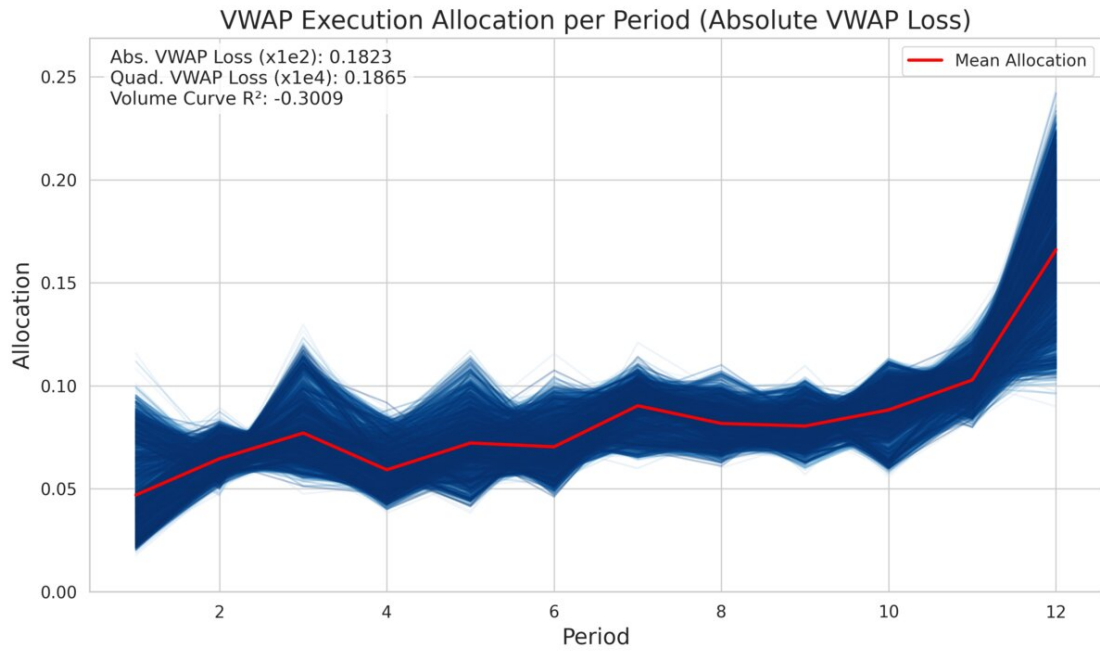


Figure 2: Prediction obtained with model calibrated using Absolute Deviation Loss

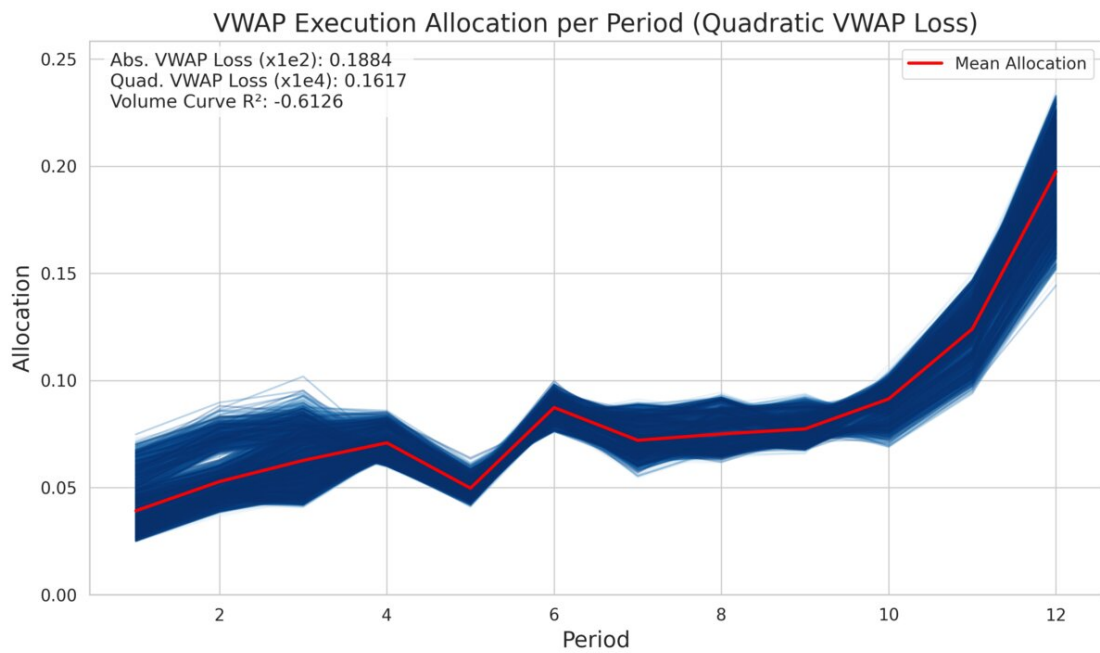


Figure 3: Prediction obtained with model calibrated using Quadratic Deviation Loss

In contrast, Figure 4 shows the allocation curve from a model trained with volume curve

loss, which is noticeably flatter and exhibits distinct seasonal waves. These differences underscore that directly optimizing the VWAP objective (via absolute or quadratic loss) yields a strategy that deviates from simply mimicking the historical volume curve.

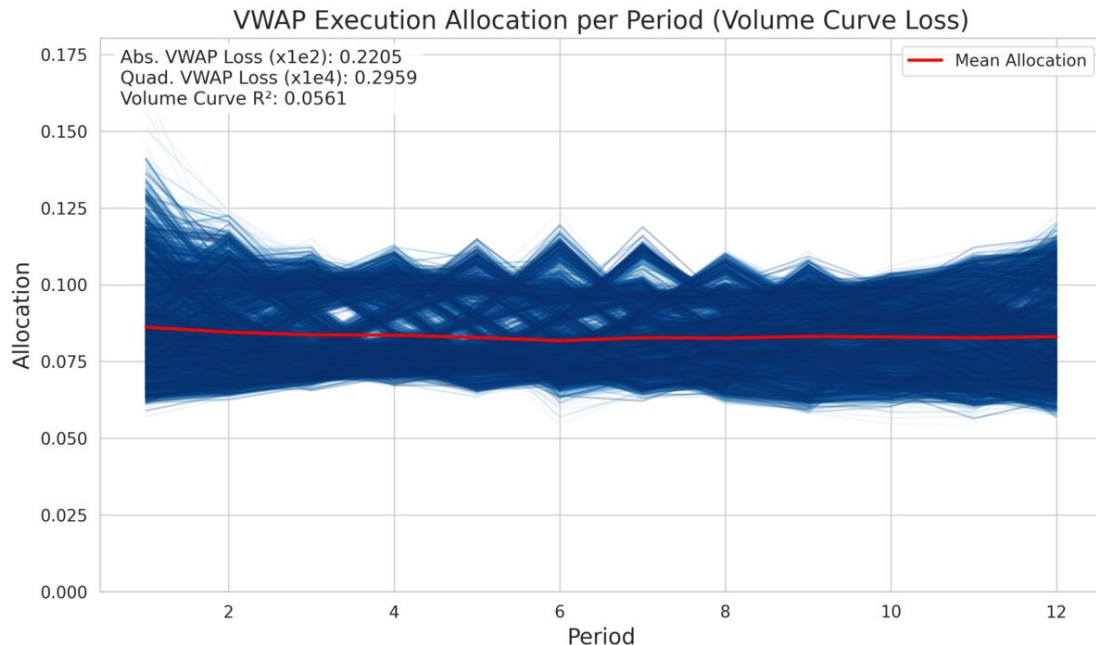


Figure 4: Prediction obtained with model calibrated using Volume Curve Loss

6.9 Understanding When It Makes a Difference

To further assess the effectiveness of the approach, an analysis of the temporal evolution of slippage (i.e., the difference between the achieved VWAP and the market VWAP) relative to the price is provided. Figure 5 shows the slippage across the full out-of-sample period for the naive approach, a dynamic VWAP strategy, and the proposed static approach when executing a 48-hour VWAP in hourly bins. Notably, the proposed method generally exhibits lower slippage during periods of high volatility, and in some instances, the slippage even reverses sign relative to standard approaches. To illustrate this effect in greater detail, Figure 6 provides a zoomed-in view over a 2000-hour subperiod. The reduced magnitude and frequency of extreme slippage events in the proposed approach are evident in this more focused view. Finally, Figures 7 and 8 present the differences in absolute slippage between the proposed method (or the dynamic linear method) and the naive approach. In these plots, negative values indicate lower absolute slippage (and thus improved performance) relative to the naive strategy, while positive values indicate higher slippage. The prevalence of negative values during volatile periods provides strong visual evidence that the end-loaded allocation strategy is effective in mitigating execution slippage.

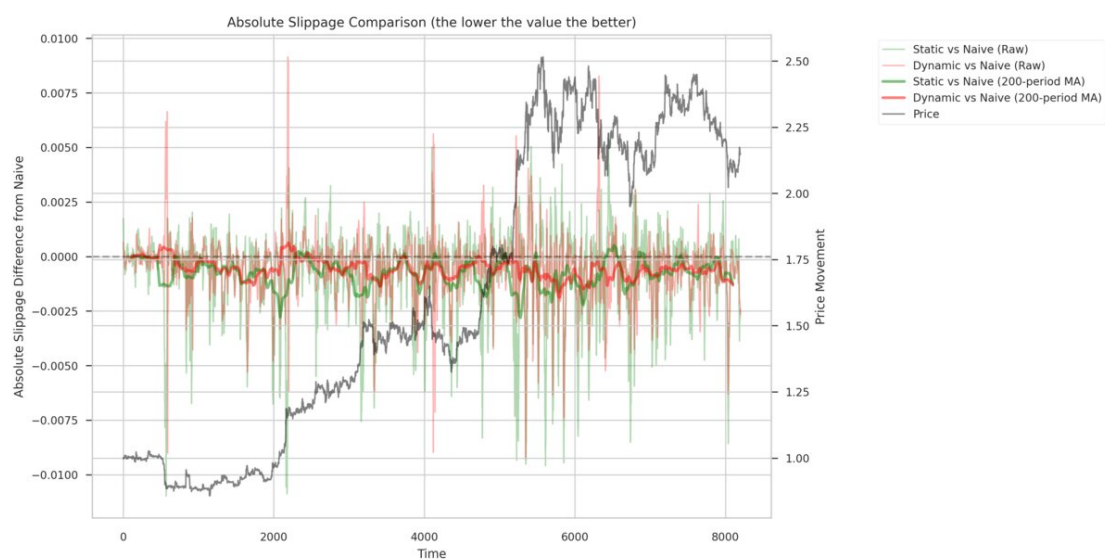


Figure 5: Slippage between approaches on the full out-of-sample set

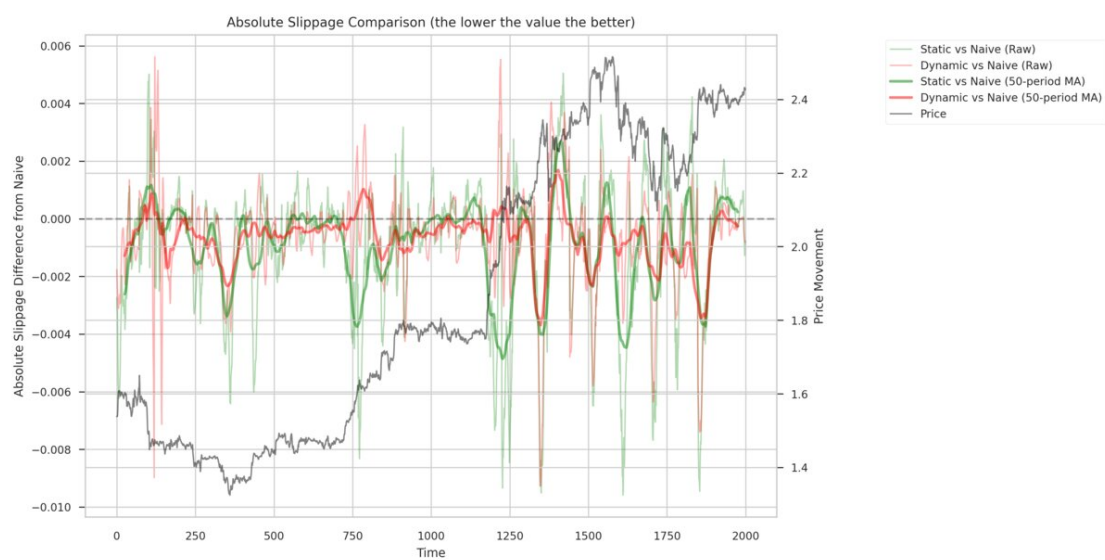


Figure 6: Slippage between approaches on a subsample of the out-of-sample set



Figure 7: Difference in absolute slippage versus naive approach on the full out-of-sample set. Negative values indicate improved performance over the naive approach.

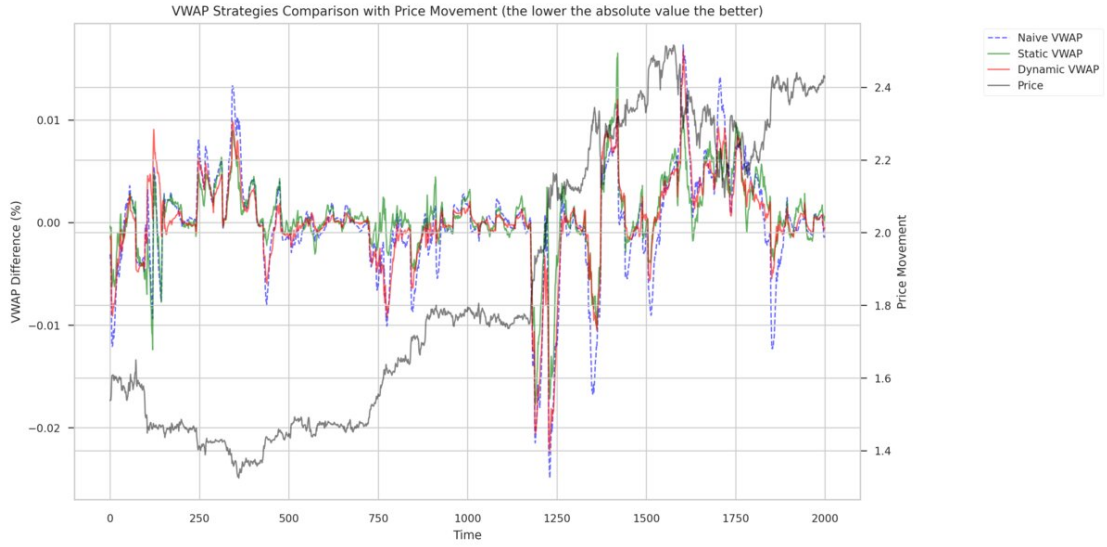


Figure 8: Difference in absolute slippage versus naive approach on the subsample set. Negative values indicate improved performance over the naive approach.

7 Conclusion

In conclusion, the results of this study demonstrate that directly optimizing the VWAP execution objective through a deep learning framework yields substantial performance

improvements over traditional volume curve-based approaches. More broadly, this example illustrates that in many cases where multi-step methods or proxy variables have traditionally been employed, deep learning offers a powerful alternative by enabling direct optimization of the true objective. The expressivity of deep learning allows for straightforward modifications to incorporate any function of market impact, thereby providing a versatile platform for capturing complex trading dynamics.

Although a simple, naive linear model (the Temporal Linear Network) was employed, its success is attributable to the fundamentally different approach of targeting the VWAP execution objective directly. This shift in methodology not only enhances execution performance but also underscores the potential of deep learning to transform similar problems in other domains. I believe that this research represents a significant advancement in the design of VWAP execution strategies, and I anticipate that further enhancements—such as dynamic updating mechanisms and the integration of sophisticated market impact models—will extend the applicability of this approach to a broader range of asset classes and trading environments.

Code Availability

The source code used for all experiments and analyses in this paper is available at <https://github.com/remigenet/DeepLearningVWAP>.

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A Results Tables

Table 3: VWAP Optimization Results for 6 steps ahead and 15 lookback window

Model Type	Asset	Optimization Function	Abs. VWAP Loss (10^{-2})		Quad. VWAP Loss (10^{-4})		R ² Vol. Curve		Training Time (s)	
			Mean	Std	Mean	Std	Mean	Std	Mean	Std
Naive	BTC	N/A	0.105363	0.000000	0.046562	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	BTC	Absolute	0.087892	0.000429	0.030610	0.000547	-0.183722	0.031797	12.667258	0.715114
StaticVWAP	BTC	Quadratic	0.089215	0.000280	0.027780	0.000445	-0.534662	0.054181	12.640691	0.591074
StaticVWAP	BTC	Volume	0.103216	0.000442	0.048582	0.000407	0.081896	0.001127	17.762622	2.642843
Dynamic Linear	BTC	N/A	0.100858	0.000000	0.048049	0.000000	0.109054	0.000000	0.049620	0.000000
Static Linear	BTC	N/A	0.102500	0.000000	0.048115	0.000000	0.087037	0.000000	0.035565	0.000000
Dynamic StaticVWAP	BTC	Absolute	0.094378	0.001417	0.038101	0.001182	-0.024425	0.016048	11.196409	0.682693
Dynamic StaticVWAP	BTC	Quadratic	0.098605	0.004904	0.041420	0.004580	-0.060975	0.034466	11.247421	0.582951
Dynamic StaticVWAP	BTC	Volume	0.102808	0.000661	0.049507	0.000656	0.095136	0.001851	14.932640	2.098674
Fixed Volume Curve	BTC	Absolute	0.090429	0.000000	0.030996	0.000000	-0.233945	0.000000	5.167805	0.000000
Fixed Volume Curve	BTC	Quadratic	0.090701	0.000000	0.028026	0.000000	-0.543176	0.000000	6.107834	0.000000
Fixed Volume Curve	BTC	Volume	0.105142	0.000000	0.046175	0.000000	0.000909	0.000000	4.947306	0.000000
Naive	ETH	N/A	0.121152	0.000000	0.061690	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	ETH	Absolute	0.100699	0.000554	0.042535	0.000771	-0.171298	0.026887	12.739415	1.228726
StaticVWAP	ETH	Quadratic	0.101188	0.000298	0.039114	0.000351	-0.438104	0.043960	12.285954	0.616164
StaticVWAP	ETH	Volume	0.117997	0.000835	0.065111	0.000817	0.073627	0.001178	16.056946	2.581432
Dynamic Linear	ETH	N/A	0.114201	0.000000	0.063881	0.000000	0.103432	0.000000	0.034144	0.000000
Static Linear	ETH	N/A	0.117400	0.000000	0.065136	0.000000	0.075838	0.000000	0.024283	0.000000
Dynamic StaticVWAP	ETH	Absolute	0.107189	0.001218	0.050199	0.001201	-0.024604	0.016778	10.827399	0.818096
Dynamic StaticVWAP	ETH	Quadratic	0.108867	0.005334	0.050609	0.005237	-0.090708	0.049720	10.975629	0.566742
Dynamic StaticVWAP	ETH	Volume	0.116336	0.001107	0.065519	0.001174	0.091956	0.002112	14.591255	2.019878
Fixed Volume Curve	ETH	Absolute	0.103074	0.000000	0.041647	0.000000	-0.235016	0.000000	5.236793	0.000000
Fixed Volume Curve	ETH	Quadratic	0.102993	0.000000	0.038466	0.000000	-0.501563	0.000000	7.602352	0.000000
Fixed Volume Curve	ETH	Volume	0.121202	0.000000	0.061560	0.000000	0.000962	0.000000	10.572813	0.000000
Naive	BNB	N/A	0.114749	0.000000	0.058204	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	BNB	Absolute	0.098716	0.000371	0.040009	0.000516	-0.179600	0.021833	11.748436	0.604826
StaticVWAP	BNB	Quadratic	0.102098	0.000456	0.037119	0.000263	-0.555685	0.036628	11.970830	0.563751
StaticVWAP	BNB	Volume	0.111265	0.000567	0.059603	0.000614	0.057254	0.000958	17.131891	2.764334
Dynamic Linear	BNB	N/A	0.109450	0.000000	0.060628	0.000000	0.083574	0.000000	0.021749	0.000000
Static Linear	BNB	N/A	0.111120	0.000000	0.060123	0.000000	0.061241	0.000000	0.021126	0.000000
Dynamic StaticVWAP	BNB	Absolute	0.100140	0.000957	0.043232	0.001259	-0.076393	0.026245	10.566655	0.743802
Dynamic StaticVWAP	BNB	Quadratic	0.102059	0.003182	0.043621	0.004242	-0.168157	0.100905	10.780092	0.561580
Dynamic StaticVWAP	BNB	Volume	0.110698	0.001120	0.061614	0.001189	0.069469	0.007095	13.911978	1.804252
Fixed Volume Curve	BNB	Absolute	0.100777	0.000000	0.040704	0.000000	-0.229291	0.000000	5.064288	0.000000
Fixed Volume Curve	BNB	Quadratic	0.102581	0.000000	0.037567	0.000000	-0.554224	0.000000	7.126949	0.000000
Fixed Volume Curve	BNB	Volume	0.113358	0.000000	0.056691	0.000000	0.000029	0.000000	3.750299	0.000000
Naive	ADA	N/A	0.152054	0.000000	0.113711	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	ADA	Absolute	0.132755	0.000331	0.081472	0.001282	-0.193195	0.020878	11.714828	0.480636
StaticVWAP	ADA	Quadratic	0.138740	0.000839	0.073484	0.000743	-0.573241	0.046742	12.054475	0.541245
StaticVWAP	ADA	Volume	0.148776	0.000583	0.122563	0.001312	0.058059	0.002759	16.160650	2.211096
Dynamic Linear	ADA	N/A	0.145723	0.000000	0.123286	0.000000	0.089187	0.000000	0.018582	0.000000
Static Linear	ADA	N/A	0.147620	0.000000	0.120900	0.000000	0.065587	0.000000	0.018901	0.000000
Dynamic StaticVWAP	ADA	Absolute	0.133511	0.000581	0.082349	0.002144	-0.156369	0.034254	10.611309	0.607574
Dynamic StaticVWAP	ADA	Quadratic	0.137696	0.004007	0.089385	0.007844	-0.134158	0.055278	10.694741	0.564848
Dynamic StaticVWAP	ADA	Volume	0.147723	0.001160	0.123852	0.001922	0.072364	0.005116	13.739881	1.852826
Fixed Volume Curve	ADA	Absolute	0.134541	0.000000	0.077918	0.000000	-0.249498	0.000000	5.076644	0.000000
Fixed Volume Curve	ADA	Quadratic	0.139507	0.000000	0.070773	0.000000	-0.683673	0.000000	6.158613	0.000000
Fixed Volume Curve	ADA	Volume	0.151185	0.000000	0.112242	0.000000	-0.000125	0.000000	5.369296	0.000000
Naive	XRP	N/A	0.144983	0.000000	0.145303	0.000000	0.000000	0.000000	0.000000	0.000000
StaticVWAP	XRP	Absolute	0.125194	0.000205	0.103814	0.001744	-0.234420	0.027725	11.967038	0.506264
StaticVWAP	XRP	Quadratic	0.131908	0.000899	0.088307	0.001263	-0.838946	0.077896	12.080394	0.594669
StaticVWAP	XRP	Volume	0.142387	0.000941	0.150526	0.001568	0.042126	0.004917	13.581942	2.835409
Dynamic Linear	XRP	N/A	0.140321	0.000000	0.150192	0.000000	0.066619	0.000000	0.019342	0.000000
Static Linear	XRP	N/A	0.141379	0.000000	0.148847	0.000000	0.047692	0.000000	0.023341	0.000000
Dynamic StaticVWAP	XRP	Absolute	0.126336	0.000554	0.102693	0.003199	-0.261795	0.057650	10.709266	0.465655
Dynamic StaticVWAP	XRP	Quadratic	0.133498	0.008056	0.101611	0.014050	-0.592432	0.456198	10.783978	0.391321
Dynamic StaticVWAP	XRP	Volume	0.142820	0.001962	0.152208	0.002334	0.047364	0.011307	11.902569	1.252472
Fixed Volume Curve	XRP	Absolute	0.126416	0.000000	0.100266	0.000000	-0.282541	0.000000	4.858923	0.000000
Fixed Volume Curve	XRP	Quadratic	0.130682	0.000000	0.084545	0.000000	-0.734930	0.000000	7.530367	0.000000
Fixed Volume Curve	XRP	Volume	0.143751	0.000000	0.142603	0.000000	0.000280	0.000000	4.148712	0.000000

Table 4: VWAP Optimization Results for 48 steps ahead and 120 lookback window

Model Type	Asset	Optimization Function	Abs. VWAP Loss (10^{-2}) Mean Std	Quad. VWAP Loss (10^{-4}) Mean Std	R ² Vol. Curve Mean Std	Training Time (s) Mean Std
Naive	BTC	N/A	0.294555 0.000000	0.240161 0.000000	0.000000 0.000000	0.000000 0.000000
StaticVWAP	BTC	Absolute	0.211433 0.004159	0.120992 0.005565	-0.164033 0.183986	14.939717 1.717610
StaticVWAP	BTC	Quadratic	0.215783 0.002679	0.119468 0.003441	-0.603772 0.287578	13.705642 0.736571
StaticVWAP	BTC	Volume	0.271076 0.004552	0.211014 0.006431	0.156645 0.007209	13.847590 0.600646
Dynamic Linear	BTC	N/A	0.255156 0.000000	0.214156 0.000000	0.259255 0.000000	0.266352 0.000000
Static Linear	BTC	N/A	0.265879 0.000000	0.206610 0.000000	0.193961 0.000000	0.262161 0.000000
Dynamic StaticVWAP	BTC	Absolute	0.315446 0.115326	0.241844 0.201273	-1.847108 3.194064	13.079563 1.249241
Dynamic StaticVWAP	BTC	Quadratic	0.444085 0.181170	0.462257 0.351248	-3.587434 4.243950	12.183085 0.628731
Dynamic StaticVWAP	BTC	Volume	0.284008 0.013071	0.234316 0.020490	0.163476 0.017579	12.310637 0.828832
Fixed Volume Curve	BTC	Absolute	0.232173 0.000000	0.139101 0.000000	-0.320558 0.000000	203.007813 0.000000
Fixed Volume Curve	BTC	Quadratic	0.223227 0.000000	0.119850 0.000000	-0.404871 0.000000	457.121906 0.000000
Fixed Volume Curve	BTC	Volume	0.299372 0.000000	0.245986 0.000000	-0.014183 0.000000	233.146574 0.000000
Naive	ETH	N/A	0.306788 0.000000	0.298487 0.000000	0.000000 0.000000	0.000000 0.000000
StaticVWAP	ETH	Absolute	0.231228 0.002613	0.175601 0.005331	-0.330961 0.214794	13.861206 0.746753
StaticVWAP	ETH	Quadratic	0.241180 0.003945	0.178887 0.007492	-0.476988 0.332822	13.535372 0.576845
StaticVWAP	ETH	Volume	0.290899 0.006168	0.290979 0.009792	0.129610 0.009962	13.552314 0.655621
Dynamic Linear	ETH	N/A	0.263310 0.000000	0.289951 0.000000	0.229208 0.000000	0.265109 0.000000
Static Linear	ETH	N/A	0.284703 0.000000	0.289589 0.000000	0.160990 0.000000	0.340382 0.000000
Dynamic StaticVWAP	ETH	Absolute	0.499737 0.177198	0.553371 0.373608	-3.896599 3.884830	12.327220 0.820315
Dynamic StaticVWAP	ETH	Quadratic	0.647119 0.234991	0.928275 0.630363	-6.944855 8.297873	11.924670 0.644082
Dynamic StaticVWAP	ETH	Volume	0.292858 0.013695	0.302941 0.022585	0.136741 0.013402	12.005558 0.575222
Fixed Volume Curve	ETH	Absolute	0.250284 0.000000	0.183056 0.000000	-0.350797 0.000000	176.584955 0.000000
Fixed Volume Curve	ETH	Quadratic	0.246691 0.000000	0.167288 0.000000	-0.577755 0.000000	482.048066 0.000000
Fixed Volume Curve	ETH	Volume	0.303970 0.000000	0.295694 0.000000	-0.019532 0.000000	146.755024 0.000000
Naive	BNB	N/A	0.306506 0.000000	0.276410 0.000000	0.000000 0.000000	0.000000 0.000000
StaticVWAP	BNB	Absolute	0.238088 0.001673	0.159082 0.002518	-0.283719 0.178850	14.012146 1.414659
StaticVWAP	BNB	Quadratic	0.274563 0.004827	0.194232 0.005945	-0.772361 0.345562	13.090553 0.609440
StaticVWAP	BNB	Volume	0.282180 0.003849	0.245898 0.006076	0.092449 0.006266	13.403553 0.599346
Dynamic Linear	BNB	N/A	0.270936 0.000000	0.245040 0.000000	0.163863 0.000000	0.244099 0.000000
Static Linear	BNB	N/A	0.279849 0.000000	0.247151 0.000000	0.109303 0.000000	0.247065 0.000000
Dynamic StaticVWAP	BNB	Absolute	0.847744 0.171242	1.565226 0.605965	-13.173198 9.265734	12.176307 1.000202
Dynamic StaticVWAP	BNB	Quadratic	0.869293 0.184199	1.674239 0.635109	-11.551874 9.018422	11.636941 0.632209
Dynamic StaticVWAP	BNB	Volume	0.290814 0.007511	0.263559 0.012282	0.103465 0.012668	11.875909 0.509390
Fixed Volume Curve	BNB	Absolute	0.259528 0.000000	0.183188 0.000000	-0.417115 0.000000	192.202802 0.000000
Fixed Volume Curve	BNB	Quadratic	0.271884 0.000000	0.188699 0.000000	-0.677903 0.000000	549.310031 0.000000
Fixed Volume Curve	BNB	Volume	0.309604 0.000000	0.280651 0.000000	-0.022514 0.000000	210.039872 0.000000
Naive	ADA	N/A	0.419024 0.000000	0.669289 0.000000	0.000000 0.000000	0.000000 0.000000
StaticVWAP	ADA	Absolute	0.351652 0.002676	0.441734 0.008008	-0.370866 0.213148	13.560816 0.722914
StaticVWAP	ADA	Quadratic	0.368340 0.004074	0.412750 0.009218	-0.734764 0.333143	13.196249 0.670882
StaticVWAP	ADA	Volume	0.408287 0.004543	0.733465 0.015243	0.084650 0.003713	13.204992 0.649402
Dynamic Linear	ADA	N/A	0.388671 0.000000	0.792817 0.000000	0.172699 0.000000	0.251263 0.000000
Static Linear	ADA	N/A	0.405453 0.000000	0.754683 0.000000	0.107097 0.000000	0.249000 0.000000
Dynamic StaticVWAP	ADA	Absolute	1.271938 0.250209	2.927095 1.148054	-17.911495 11.084118	11.941535 0.699107
Dynamic StaticVWAP	ADA	Quadratic	1.290262 0.241712	2.988568 1.031134	-14.352830 7.704632	11.799472 0.598161
Dynamic StaticVWAP	ADA	Volume	0.421940 0.009454	0.769364 0.033799	0.086176 0.011173	11.787223 0.514459
Fixed Volume Curve	ADA	Absolute	0.353334 0.000000	0.401683 0.000000	-0.517208 0.000000	197.274881 0.000000
Fixed Volume Curve	ADA	Quadratic	0.373185 0.000000	0.381716 0.000000	-0.777649 0.000000	484.393773 0.000000
Fixed Volume Curve	ADA	Volume	0.412232 0.000000	0.646860 0.000000	-0.019907 0.000000	224.401588 0.000000
Naive	XRP	N/A	0.417425 0.000000	0.652992 0.000000	0.000000 0.000000	0.000000 0.000000
StaticVWAP	XRP	Absolute	0.336897 0.002525	0.394574 0.012198	-0.385838 0.207828	13.638464 0.637926
StaticVWAP	XRP	Quadratic	0.366798 0.008950	0.338236 0.015288	-1.488659 0.668829	13.407381 0.678238
StaticVWAP	XRP	Volume	0.405872 0.003963	0.670118 0.009434	0.062685 0.005969	13.312675 0.646472
Dynamic Linear	XRP	N/A	0.391767 0.000000	0.675384 0.000000	0.135234 0.000000	0.237720 0.000000
Static Linear	XRP	N/A	0.400026 0.000000	0.669650 0.000000	0.077930 0.000000	0.268407 0.000000
Dynamic StaticVWAP	XRP	Absolute	0.829306 0.172708	1.305078 0.532350	-11.319136 7.766351	12.344783 1.181881
Dynamic StaticVWAP	XRP	Quadratic	0.744789 0.220680	1.118644 0.615147	-8.111042 7.154984	12.244301 1.173284
Dynamic StaticVWAP	XRP	Volume	0.416765 0.010058	0.707899 0.030212	0.071229 0.008034	11.917936 0.555746
Fixed Volume Curve	XRP	Absolute	0.344144 0.000000	0.380628 0.000000	-0.411449 0.000000	195.540658 0.000000
Fixed Volume Curve	XRP	Quadratic	0.357687 0.000000	0.328731 0.000000	-0.623001 0.000000	559.380669 0.000000
Fixed Volume Curve	XRP	Volume	0.414514 0.000000	0.642567 0.000000	-0.015077 0.000000	211.496234 0.000000

B Execution Curves Graphs

B.1 BTC Execution Curves with 6 Steps Ahead

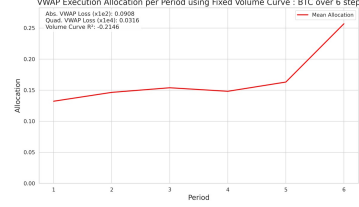


Figure 9: Fixed Volume Curve (Absolute Loss)

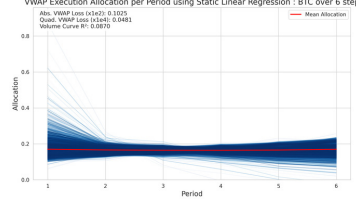


Figure 12: Static Linear Regression

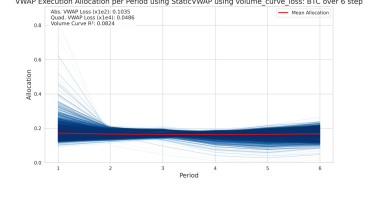


Figure 16: StaticVWAP (Volume Curve Loss)

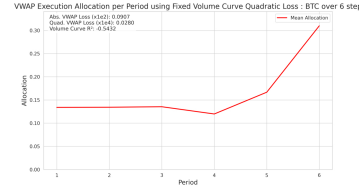


Figure 10: Fixed Volume Curve (Quadratic Loss)

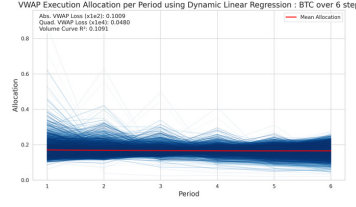


Figure 13: Dynamic Linear Regression

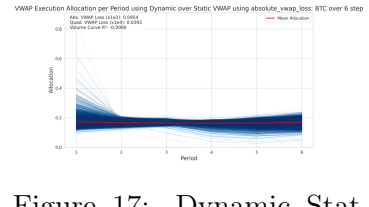


Figure 17: Dynamic StaticVWAP (Absolute Loss)

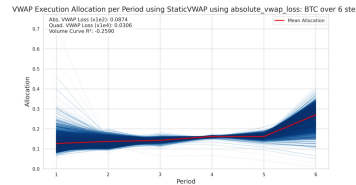


Figure 14: StaticVWAP (Absolute Loss)

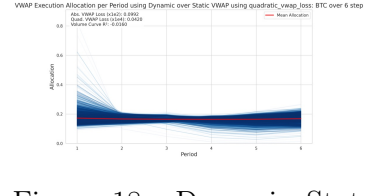


Figure 18: Dynamic StaticVWAP (Quadratic Loss)

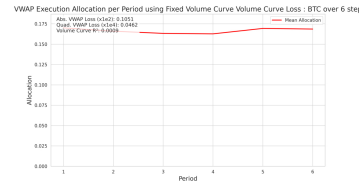


Figure 11: Fixed Volume Curve (Volume Curve Loss)

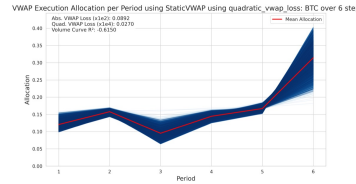


Figure 15: StaticVWAP (Quadratic Loss)

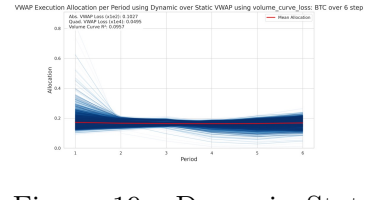


Figure 19: Dynamic StaticVWAP (Volume Curve Loss)

B.2 ETH Execution Curves with 6 Steps Ahead

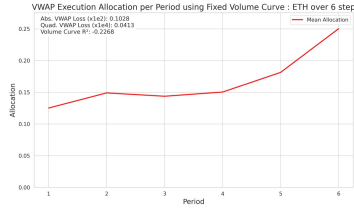


Figure 20: Fixed Volume Curve (Absolute Loss)

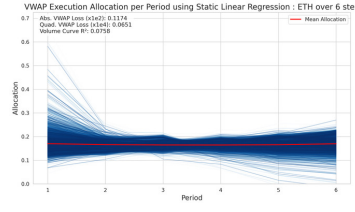


Figure 23: Static Linear Regression

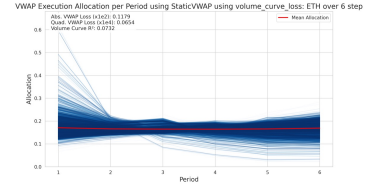


Figure 27: StaticVWAP (Volume Curve Loss)

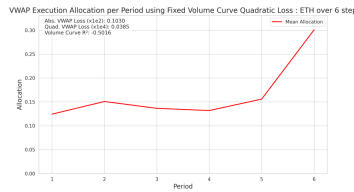


Figure 21: Fixed Volume Curve (Quadratic Loss)

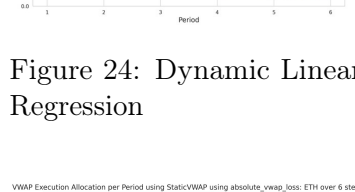


Figure 24: Dynamic Linear Regression

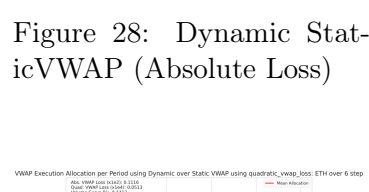


Figure 28: Dynamic StaticVWAP (Absolute Loss)

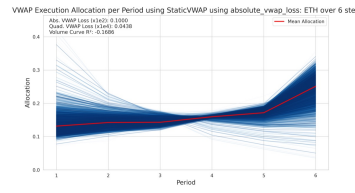


Figure 25: StaticVWAP (Absolute Loss)

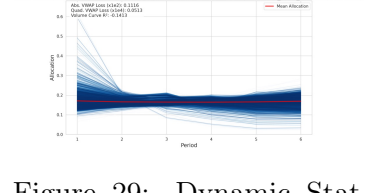


Figure 29: Dynamic StaticVWAP (Quadratic Loss)

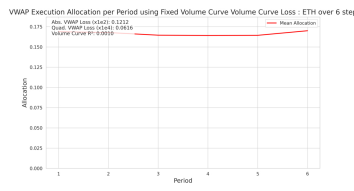


Figure 22: Fixed Volume Curve (Volume Curve Loss)

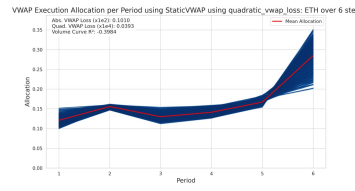


Figure 26: StaticVWAP (Quadratic Loss)

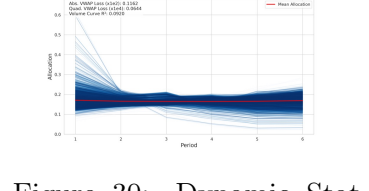


Figure 30: Dynamic StaticVWAP (Volume Curve Loss)

B.3 BNB Execution Curves with 6 Steps Ahead

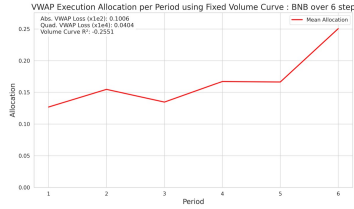


Figure 31: Fixed Volume Curve (Absolute Loss)

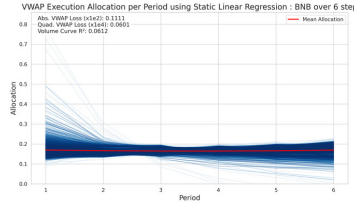


Figure 34: Static Linear Regression

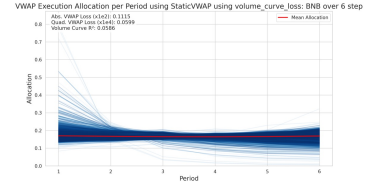


Figure 38: StaticVWAP (Volume Curve Loss)

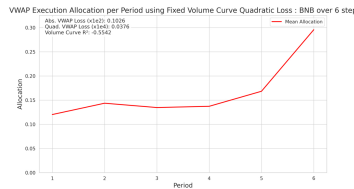


Figure 32: Fixed Volume Curve (Quadratic Loss)

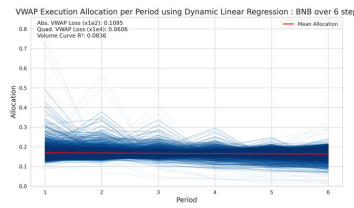


Figure 35: Dynamic Linear Regression

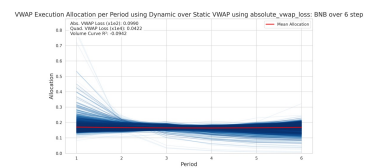


Figure 39: Dynamic StaticVWAP (Absolute Loss)

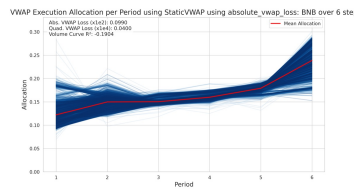


Figure 36: StaticVWAP (Absolute Loss)

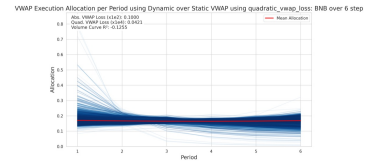


Figure 40: Dynamic StaticVWAP (Quadratic Loss)

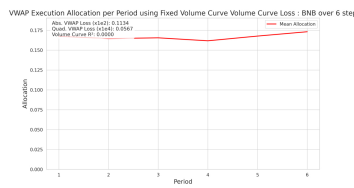


Figure 33: Fixed Volume Curve (Volume Curve Loss)

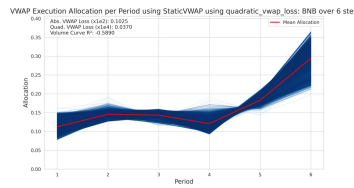


Figure 37: StaticVWAP (Quadratic Loss)

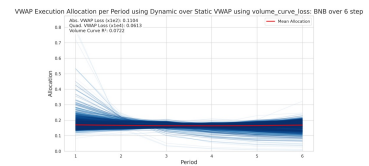


Figure 41: Dynamic StaticVWAP (Volume Curve Loss)

B.4 ADA Execution Curves with 6 Steps Ahead

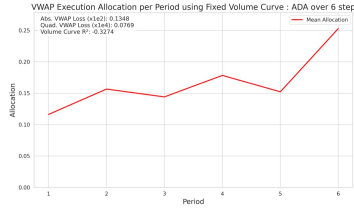


Figure 42: Fixed Volume Curve (Absolute Loss)

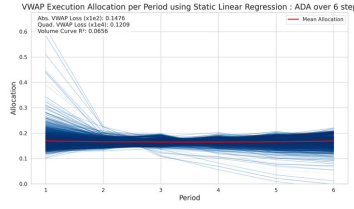


Figure 45: Static Linear Regression

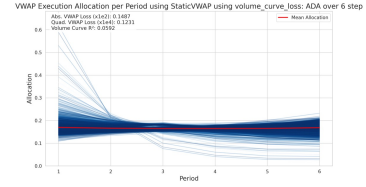


Figure 49: StaticVWAP (Volume Curve Loss)

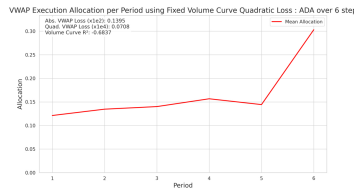


Figure 43: Fixed Volume Curve (Quadratic Loss)

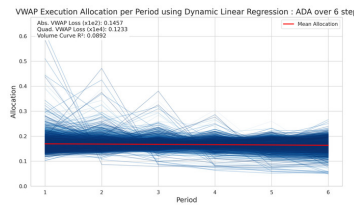


Figure 46: Dynamic Linear Regression

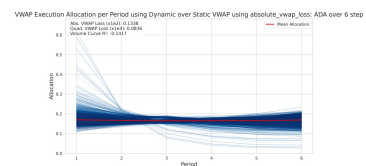


Figure 50: Dynamic StaticVWAP (Absolute Loss)

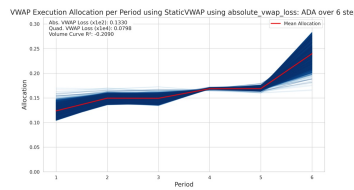


Figure 47: StaticVWAP (Absolute Loss)

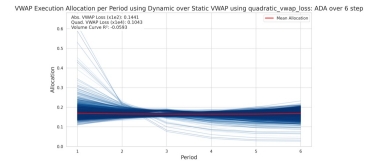


Figure 51: Dynamic StaticVWAP (Quadratic Loss)

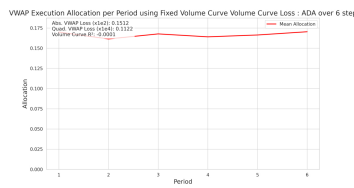


Figure 44: Fixed Volume Curve (Volume Curve Loss)

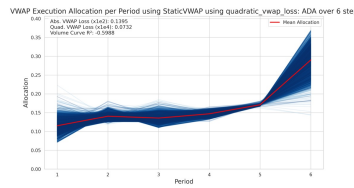


Figure 48: StaticVWAP (Quadratic Loss)

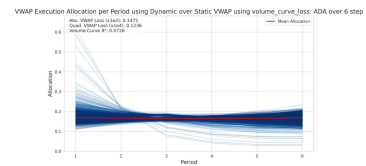


Figure 52: Dynamic StaticVWAP (Volume Curve Loss)

B.5 XRP Execution Curves with 6 Steps Ahead

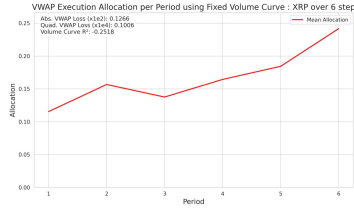


Figure 53: Fixed Volume Curve (Absolute Loss)

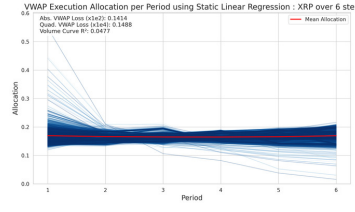


Figure 56: Static Linear Regression

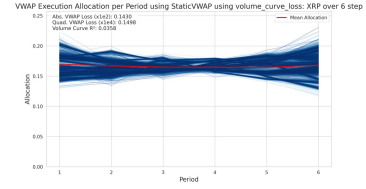


Figure 60: StaticVWAP (Volume Curve Loss)

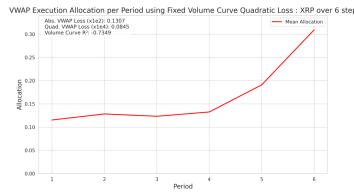


Figure 54: Fixed Volume Curve (Quadratic Loss)

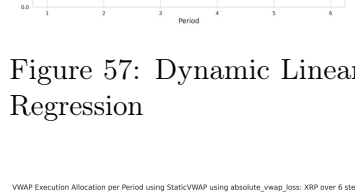


Figure 57: Dynamic Linear Regression

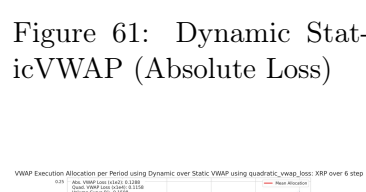


Figure 61: Dynamic StaticVWAP (Absolute Loss)

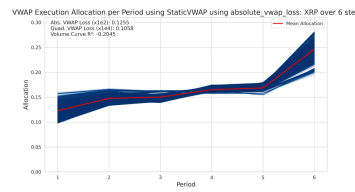


Figure 58: StaticVWAP (Absolute Loss)

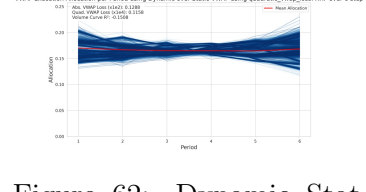


Figure 62: Dynamic StaticVWAP (Quadratic Loss)

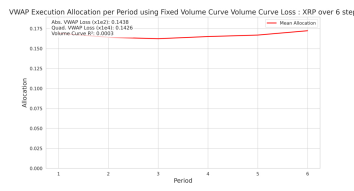


Figure 55: Fixed Volume Curve (Volume Curve Loss)

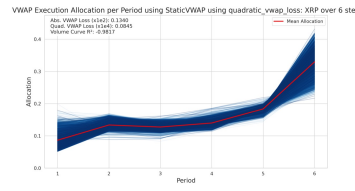


Figure 59: StaticVWAP (Quadratic Loss)

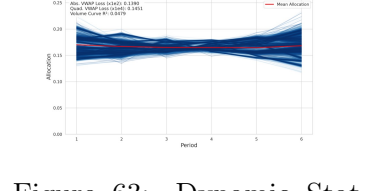


Figure 63: Dynamic StaticVWAP (Volume Curve Loss)

B.6 BTC Execution Curves with 12 Steps Ahead

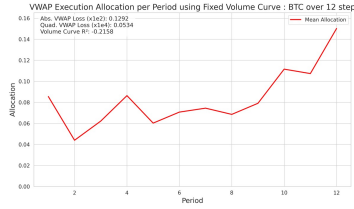


Figure 64: Fixed Volume Curve (Absolute Loss)

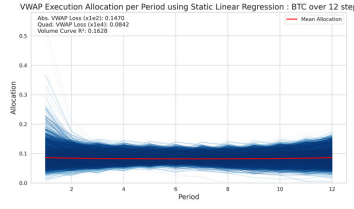


Figure 67: Static Linear Regression

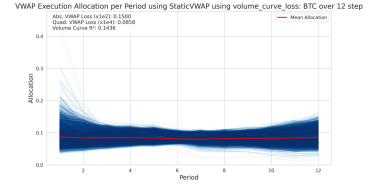


Figure 71: StaticVWAP (Volume Curve Loss)

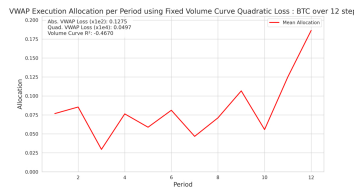


Figure 65: Fixed Volume Curve (Quadratic Loss)

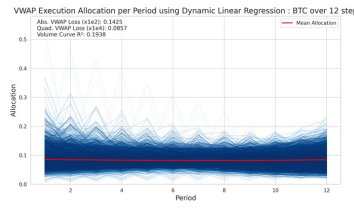


Figure 68: Dynamic Linear Regression

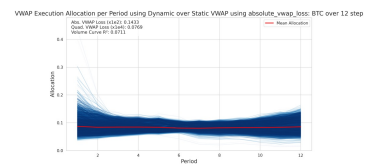


Figure 72: Dynamic StaticVWAP (Absolute Loss)

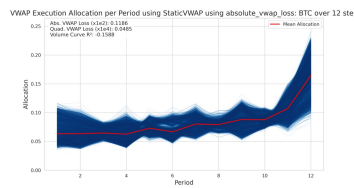


Figure 69: StaticVWAP (Absolute Loss)

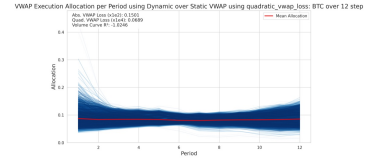


Figure 73: Dynamic StaticVWAP (Quadratic Loss)

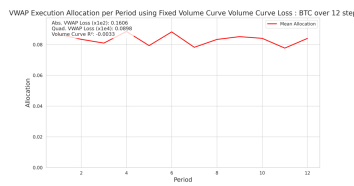


Figure 66: Fixed Volume Curve (Volume Curve Loss)

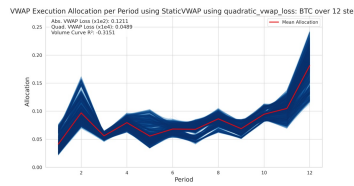


Figure 70: StaticVWAP (Quadratic Loss)

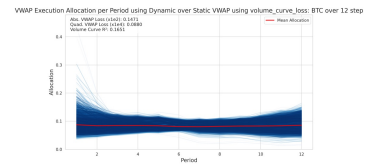


Figure 74: Dynamic StaticVWAP (Volume Curve Loss)

B.7 ETH Execution Curves with 12 Steps Ahead

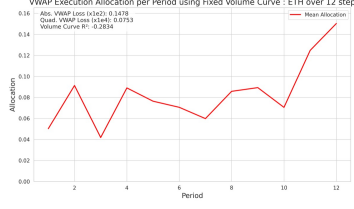


Figure 75: Fixed Volume Curve (Absolute Loss)

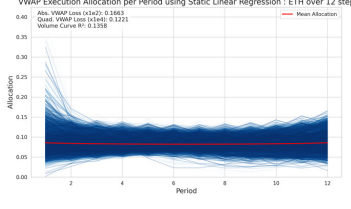


Figure 78: Static Linear Regression

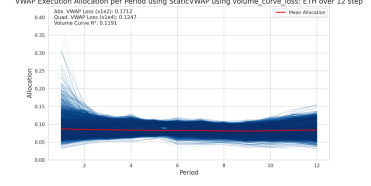


Figure 82: StaticVWAP (Volume Curve Loss)

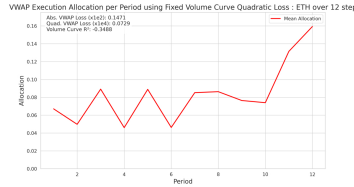


Figure 76: Fixed Volume Curve (Quadratic Loss)

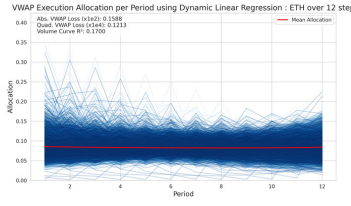


Figure 79: Dynamic Linear Regression

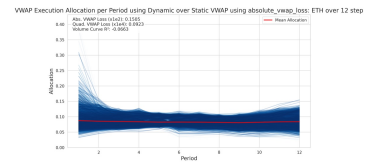


Figure 83: Dynamic StaticVWAP (Absolute Loss)

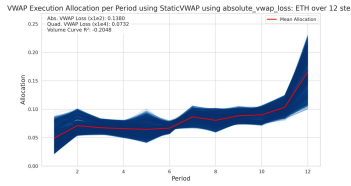


Figure 80: StaticVWAP (Absolute Loss)

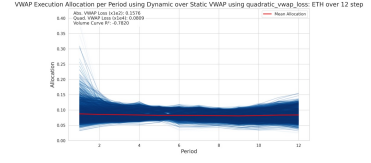


Figure 84: Dynamic StaticVWAP (Quadratic Loss)

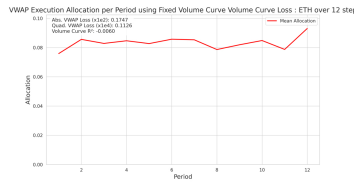


Figure 77: Fixed Volume Curve (Volume Curve Loss)

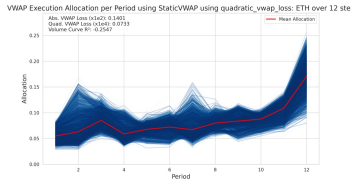


Figure 81: StaticVWAP (Quadratic Loss)

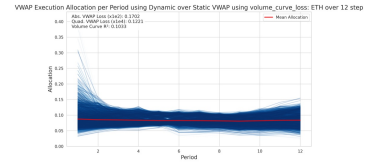


Figure 85: Dynamic StaticVWAP (Volume Curve Loss)

B.8 BNB Execution Curves with 12 Steps Ahead

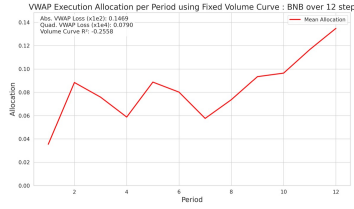


Figure 86: Fixed Volume Curve (Absolute Loss)

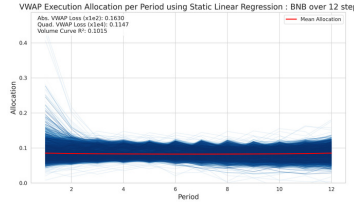


Figure 89: Static Linear Regression

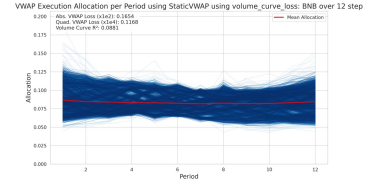


Figure 93: StaticVWAP (Volume Curve Loss)

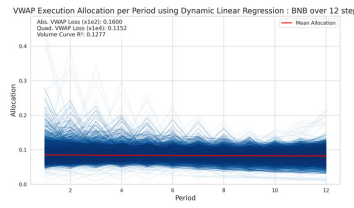


Figure 90: Dynamic Linear Regression

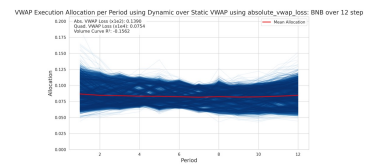


Figure 94: Dynamic StaticVWAP (Absolute Loss)

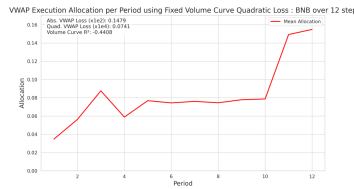


Figure 87: Fixed Volume Curve (Quadratic Loss)

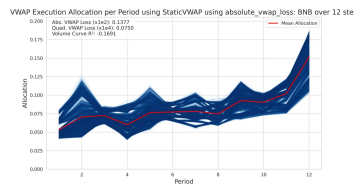


Figure 91: StaticVWAP (Absolute Loss)

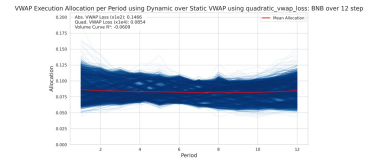


Figure 95: Dynamic StaticVWAP (Quadratic Loss)

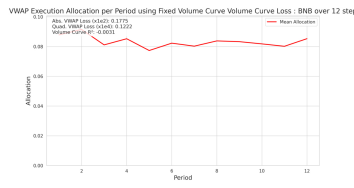


Figure 88: Fixed Volume Curve (Volume Curve Loss)

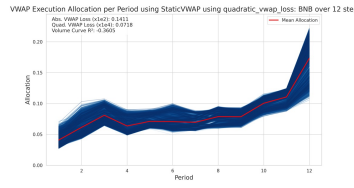


Figure 92: StaticVWAP (Quadratic Loss)

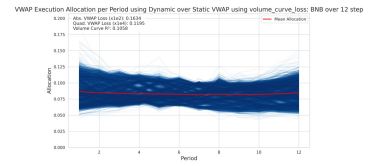


Figure 96: Dynamic StaticVWAP (Volume Curve Loss)

B.9 ADA Execution Curves with 12 Steps Ahead

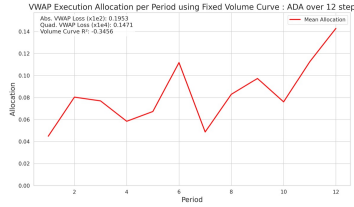


Figure 97: Fixed Volume Curve (Absolute Loss)

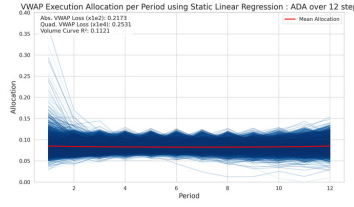


Figure 100: Static Linear Regression

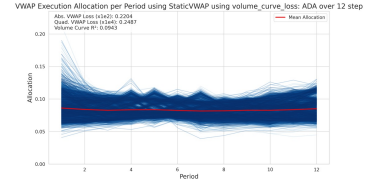


Figure 104: StaticVWAP (Volume Curve Loss)

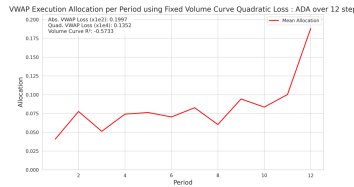


Figure 98: Fixed Volume Curve (Quadratic Loss)

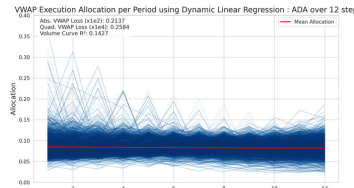


Figure 101: Dynamic Linear Regression

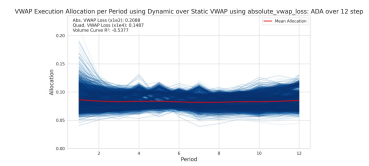


Figure 105: Dynamic StaticVWAP (Absolute Loss)

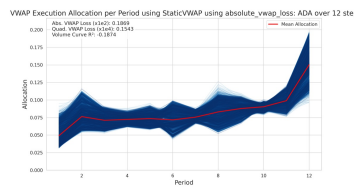


Figure 102: StaticVWAP (Absolute Loss)

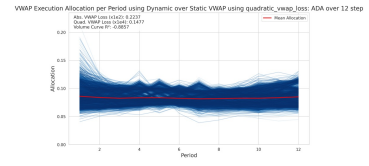


Figure 106: Dynamic StaticVWAP (Quadratic Loss)

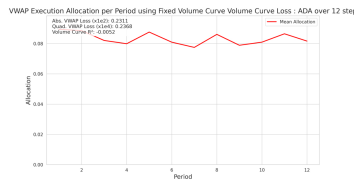


Figure 99: Fixed Volume Curve (Volume Curve Loss)

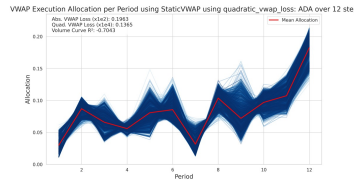


Figure 103: StaticVWAP (Quadratic Loss)

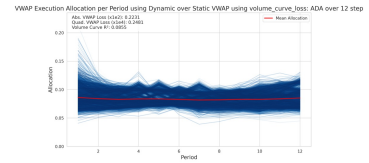


Figure 107: Dynamic StaticVWAP (Volume Curve Loss)

B.10 XRP Execution Curves with 12 Steps Ahead

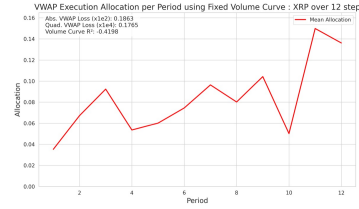


Figure 108: Fixed Volume Curve (Absolute Loss)

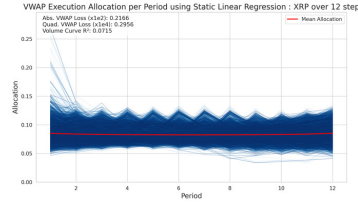


Figure 111: Static Linear Regression

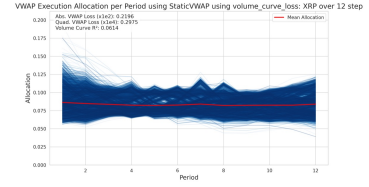


Figure 115: StaticVWAP (Volume Curve Loss)

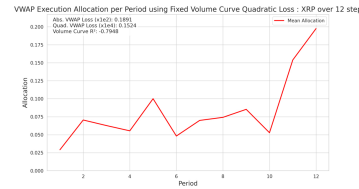


Figure 109: Fixed Volume Curve (Quadratic Loss)

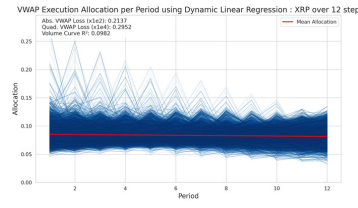


Figure 112: Dynamic Linear Regression

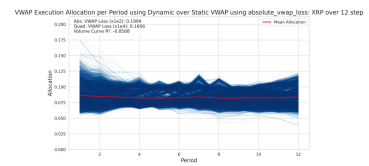


Figure 116: Dynamic StaticVWAP (Absolute Loss)

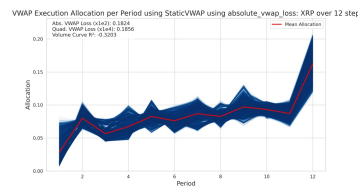


Figure 113: StaticVWAP (Absolute Loss)

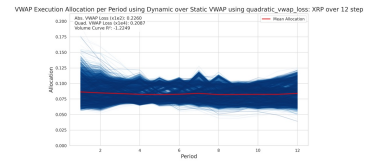


Figure 117: Dynamic StaticVWAP (Quadratic Loss)

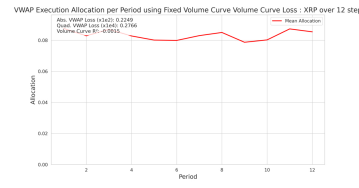


Figure 110: Fixed Volume Curve (Volume Curve Loss)

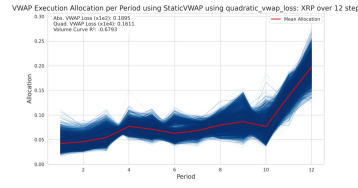


Figure 114: StaticVWAP (Quadratic Loss)

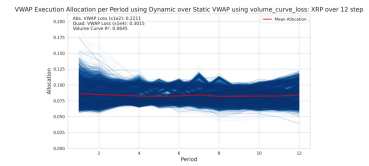


Figure 118: Dynamic StaticVWAP (Volume Curve Loss)

B.11 BTC Execution Curves with 48 Steps Ahead

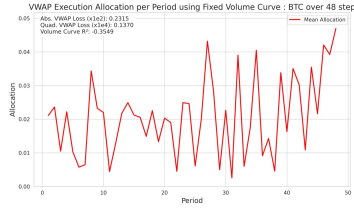


Figure 119: Fixed Volume Curve (Absolute Loss)

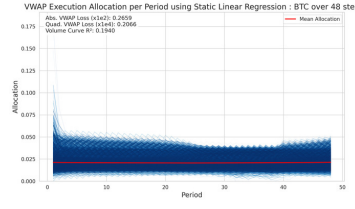


Figure 122: Static Linear Regression

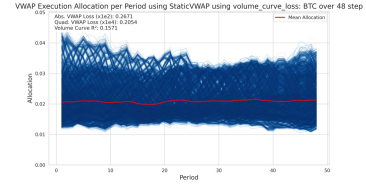


Figure 126: StaticVWAP (Volume Curve Loss)

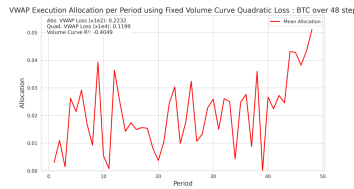


Figure 120: Fixed Volume Curve (Quadratic Loss)

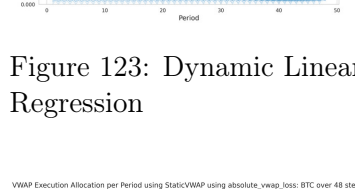


Figure 123: Dynamic Linear Regression

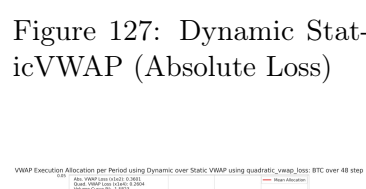


Figure 127: Dynamic StaticVWAP (Absolute Loss)

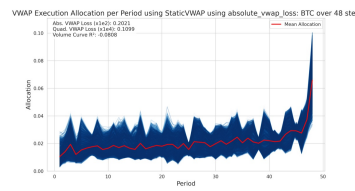


Figure 124: StaticVWAP (Absolute Loss)

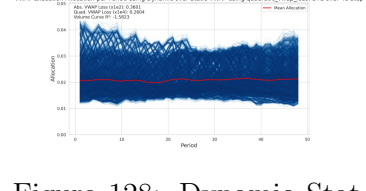


Figure 128: Dynamic StaticVWAP (Quadratic Loss)

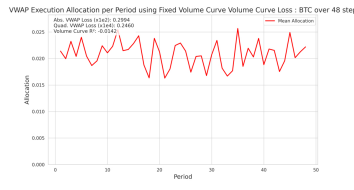


Figure 121: Fixed Volume Curve (Volume Curve Loss)

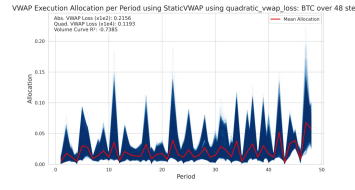


Figure 125: StaticVWAP (Quadratic Loss)

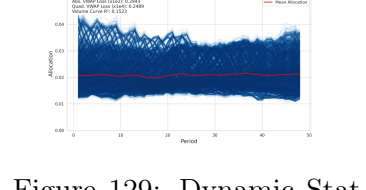


Figure 129: Dynamic StaticVWAP (Volume Curve Loss)

B.12 ETH Execution Curves with 48 Steps Ahead

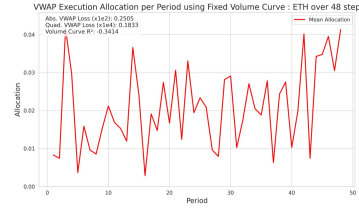


Figure 130: Fixed Volume Curve (Absolute Loss)

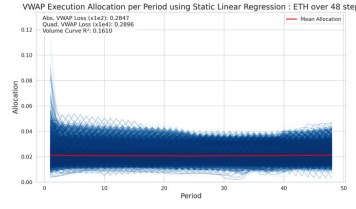


Figure 133: Static Linear Regression

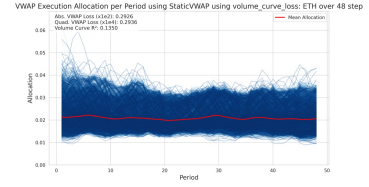


Figure 137: StaticVWAP (Volume Curve Loss)

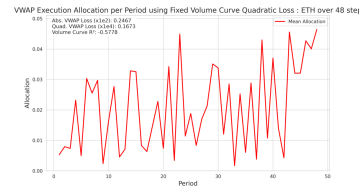


Figure 131: Fixed Volume Curve (Quadratic Loss)

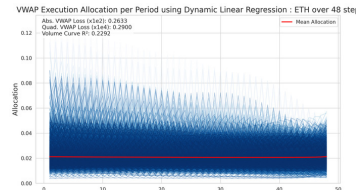


Figure 134: Dynamic Linear Regression

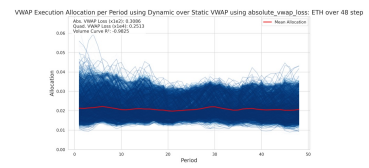


Figure 138: Dynamic StaticVWAP (Absolute Loss)

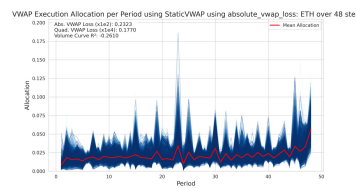


Figure 135: StaticVWAP (Absolute Loss)

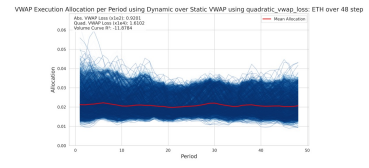


Figure 139: Dynamic StaticVWAP (Quadratic Loss)

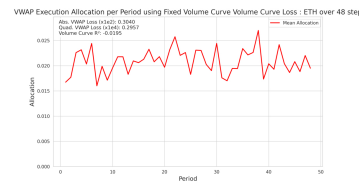


Figure 132: Fixed Volume Curve (Volume Curve Loss)

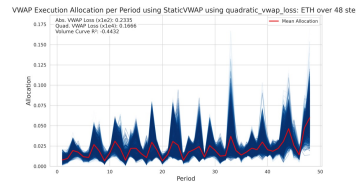


Figure 136: StaticVWAP (Quadratic Loss)

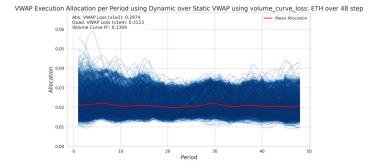


Figure 140: Dynamic StaticVWAP (Volume Curve Loss)

B.13 BNB Execution Curves with 48 Steps Ahead

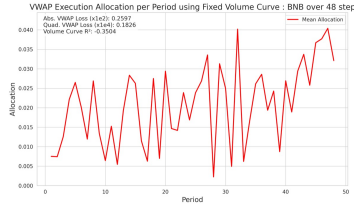


Figure 141: Fixed Volume Curve (Absolute Loss)

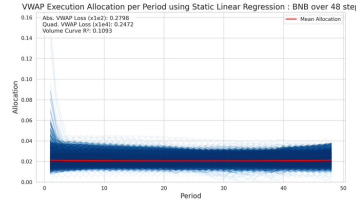


Figure 144: Static Linear Regression

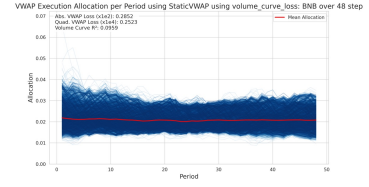


Figure 148: StaticVWAP (Volume Curve Loss)

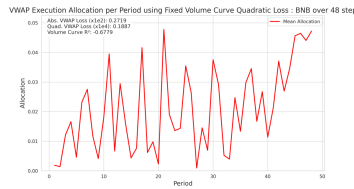


Figure 142: Fixed Volume Curve (Quadratic Loss)

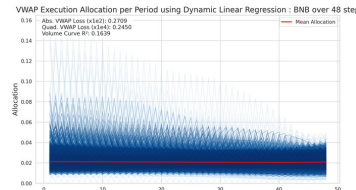


Figure 145: Dynamic Linear Regression

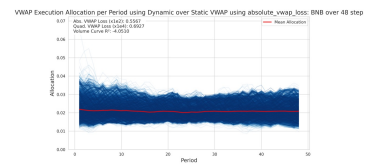


Figure 149: Dynamic StaticVWAP (Absolute Loss)

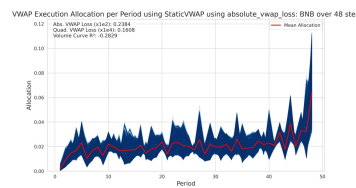


Figure 146: StaticVWAP (Absolute Loss)

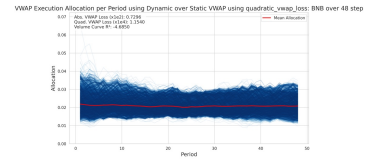


Figure 150: Dynamic StaticVWAP (Quadratic Loss)

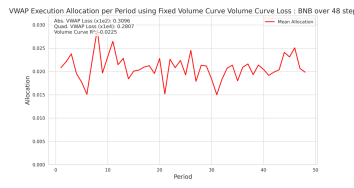


Figure 143: Fixed Volume Curve (Volume Curve Loss)

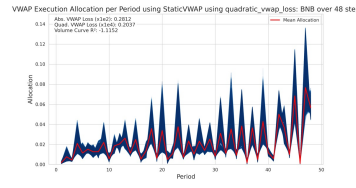


Figure 147: StaticVWAP (Quadratic Loss)

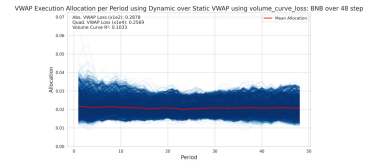


Figure 151: Dynamic StaticVWAP (Volume Curve Loss)

B.14 ADA Execution Curves with 48 Steps Ahead

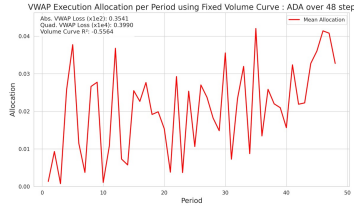


Figure 152: Fixed Volume Curve (Absolute Loss)

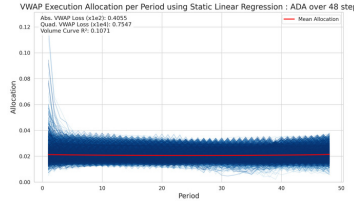


Figure 155: Static Linear Regression

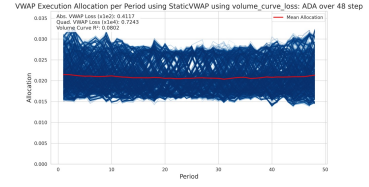


Figure 159: StaticVWAP (Volume Curve Loss)

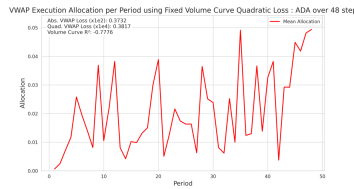


Figure 153: Fixed Volume Curve (Quadratic Loss)

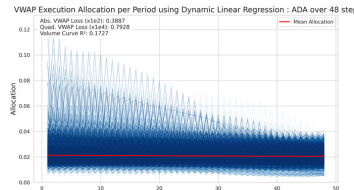


Figure 156: Dynamic Linear Regression

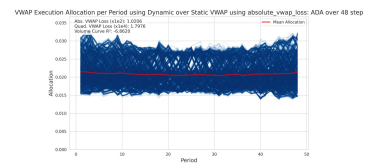


Figure 160: Dynamic StaticVWAP (Absolute Loss)

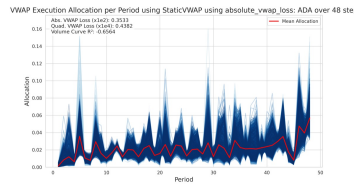


Figure 157: StaticVWAP (Absolute Loss)

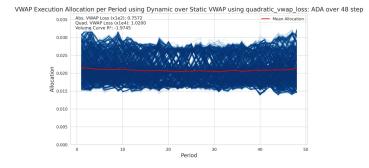


Figure 161: Dynamic StaticVWAP (Quadratic Loss)

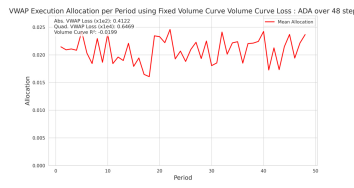


Figure 154: Fixed Volume Curve (Volume Curve Loss)

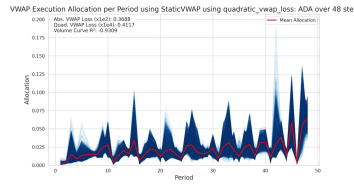


Figure 158: StaticVWAP (Quadratic Loss)

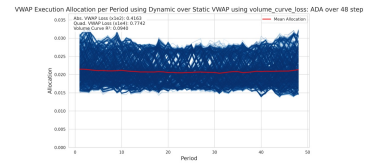


Figure 162: Dynamic StaticVWAP (Volume Curve Loss)

B.15 XRP Execution Curves with 48 Steps Ahead

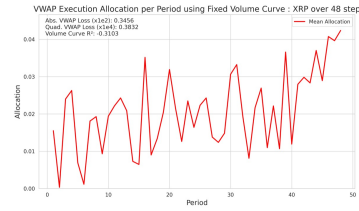


Figure 163: Fixed Volume Curve (Absolute Loss)

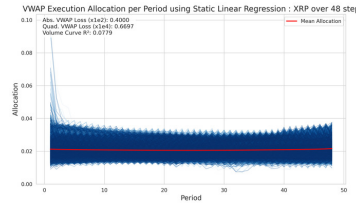


Figure 166: Static Linear Regression

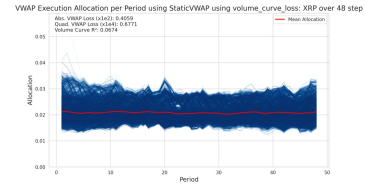


Figure 170: StaticVWAP (Volume Curve Loss)

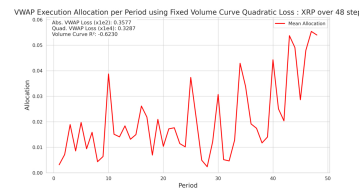


Figure 164: Fixed Volume Curve (Quadratic Loss)

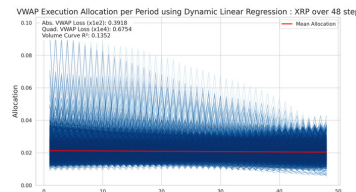


Figure 167: Dynamic Linear Regression

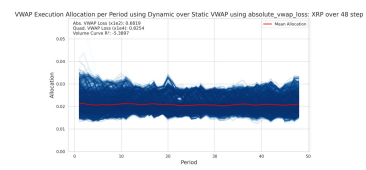


Figure 171: Dynamic StaticVWAP (Absolute Loss)

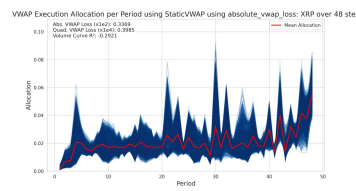


Figure 168: StaticVWAP (Absolute Loss)

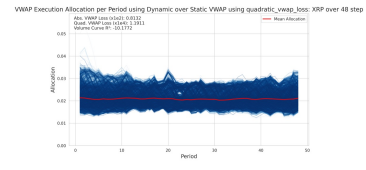


Figure 172: Dynamic StaticVWAP (Quadratic Loss)

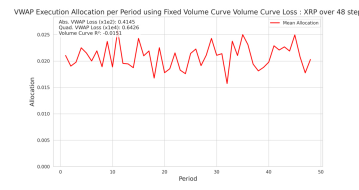


Figure 165: Fixed Volume Curve (Volume Curve Loss)

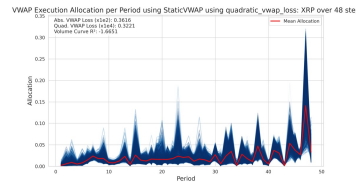


Figure 169: StaticVWAP (Quadratic Loss)

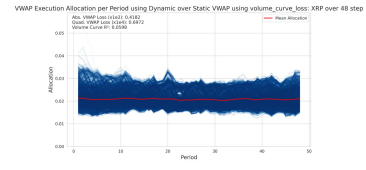


Figure 173: Dynamic StaticVWAP (Volume Curve Loss)